TFPT — Frontier items

η_B, m_p/m_e, the Koide relation, dark matter and quantum gravity — the honest status, not forced onto the ladder

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What this document is about — the status authority for the frontier

The honest frontier: which physics has a genuine TFPT handle and which does not. For each of η_B, m_p/m_e, the Koide relation, dark matter and full quantum gravity, this note states the genuine structural handle, the precision with which it currently lands, and — crucially — what is *not* a clean compiler power and is deliberately *not* forced onto the ladder. **This document is the status authority for the frontier items:** where any other document's wording and the markers here disagree, the typing here (and the `status_ledger.csv`) wins.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8, 4, 4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified).

You are reading document 4 (Frontier).

TFPT status at a glance — read this card the same way in every document

Two structural interfaces, one compiler. P1 is a primitive reflection-positive seam kernel with unit winding; its normalization is $c_3 = \frac{1}{8\pi}$. From this datum and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2), the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The numerical readouts c_3 and g_{car} are elementary symmetric readouts of the anchor $a = (1, 1, 2)$. The algebraic core is machine-checked; physical readouts run

through *named bridges* whose status is typed below. Existing reductions share the four-mark geometry and anchor $a = (1, 1, 2)$; both W-bridge premises are derived at finite level (`v1014/v1015`), modulo the external MMST identification. The additional P1 determinant-response identification is separate. The Architecture chapter "Shared origin" consolidates these scopes (`REC20261105C3`).

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725)	closed metrology [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

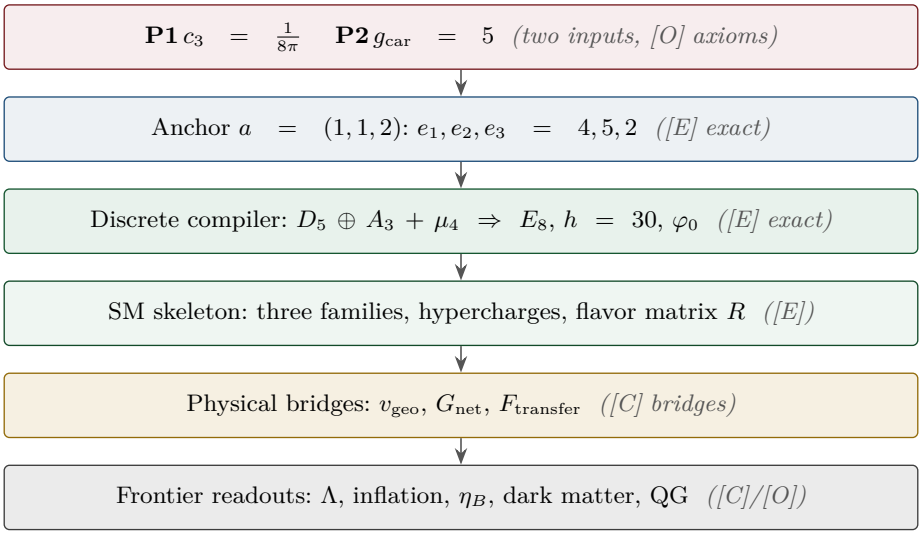
Beside that compiler rest sits the strict-physical-TOE rest: the ten-component Rest_{TOE} / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} :=$ AND(T1..T8); not a closure).

The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2) e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. The lattice and index witnesses and the cited scaling and OS theorems establish the stated conditional route; they do not leave every old prerequisite untouched. The later charged MMST manuscript proves its uniform Green, edge graph, crossed-product and relaxed TEL-B estimates and retains the named microscopic local-algebra exhaustion condition for the full source. Its P1 corollary separately requires identification of the determinant-line response with the additive transitive four-channel response. The declared operational seed already supplies state and time data. The physical question concerns applicability of these theorems to the same source. The unconditional seam parent and T1–T8 keep their ledger statuses. Under the conformal-net hypotheses, rotation invariance makes the geometric deck a consequence (`v335`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76/v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test



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Ground rule. These five are exactly the items previously flagged "*not on a clean action/ladder yet — do not force.*" This note gives, for each, the *genuine* structural handle that does exist, the precision with which it currently lands, and a clear statement of what is *not* a single compiler power. No fabricated ladder rungs; the honest markers [E]/[C]/[O] are applied throughout. (All numbers machine-verified.)

Hard rule on the frontier — one interface F_{transfer}

Koide, η_B, the axion relic scale and m_p/m_e are *not* compiler powers unless their missing QFT-transfer / cosmology computation is explicitly supplied. They are deliberately typed interfaces, not structural holes, and they are *four instances of one missing functor* F_{transfer} — the continuous transport from compiler *source* data to measured observables (Koide source→pole; η_B source→Boltzmann relic; axion scale→cosmological abundance; $m_p/m_e \to$ QCD/EW matching). Each has a genuine handle (a near-value, a determined readout, a fixed candidate) but its *exact* status is conditional on that named transfer calculation — now a runnable [C] solver with a kill test (v371–v375), still a transfer target, never a compiler power. In particular η_B is a *transfer target*, not a primitive

compiler prediction. They must never be quoted as primitive TFPT outputs.

Why this is principled, not a gap: these are precisely the **discrete↔ continuous boundary** of the theory — the points where the discrete compiler hands its skeleton off to continuous physics (QCD confinement, cosmological Boltzmann transport, relic integrals). The cleanness of a quantity tracks how far it sits from that handoff: pure compiler ratios are exact, QCD/cosmology-dressed quantities are typed interfaces. [`verification/v35_quark_qcd_boundary.py`]

Machine-enforced. This rule is a CI-style guard ([`verification/v187_ftransfer_laws.py`]): the four frontier families (F_{pole} Koide, $F_{\text{Boltzmann}}$ η_B, F_{relic} axion, F_{QCD} m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — exact algebraic sub-parts (the Koide 53/54 factor, $b_3 = -7$) may be [E], but the *physical* prediction never is. No future edit can quietly promote a frontier number to a closed compiler output. A companion *prose* sentinel ([`verification/v188_frontier_wording_guard.py`]) forbids stale sentences whose semantics an earlier round superseded, so a frozen release can never re-introduce wording that was true yesterday and is half-true today.

The functor contract `CONTRACT.F.01`. Beyond the typing guard, the functor is pinned by four checkable structural axioms ([`verification/v213_ftransfer_functor.py`]): each instance is a *discrete* compiler kernel [E] composed with an *external* continuous solver [C], and the functor is (1) μ_4-deck equivariant (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ *is* the deck transfer eigenvalue), (2) Plücker-preserving ($53 = a^T(R{+}Q)\mathbf{1}$, $\|\mathrm{Pl}(K)\|_1 = 11$), (3) positive/stochastic ($\operatorname{spec} T = \{1, (2/3)^6, (1/3)^6\}$, $\kappa_f \in (0,1)$), and (4) external-module-explicit ($b_3 = -7$ a carrier output, the Boltzmann/relic/QCD content named and external). So F_{transfer} is a typed functor, not a bag of open topics — the third research contract alongside (U_{wall}) and (G_{metric}), and a consolidation, not a closure. [2026-08-05: the jet route of the contract is closed at the physical-clock level — the preregistered external-clock kill fired sympy-exactly ([`verification/v777_ftransfer_clock_jets.py`], verdict NO-COMMON-CONNECTION: transported Schwarzians $\{-\Delta^2/2, -\Delta^2/2, 0\}$), and the surviving architecture is the *fibered* functor — one discrete PGL_2 base with the constant seam obstruction cocycle $\{0, \pm\Delta^2/2\}$; the four observables stay [C]/[O]. 2026-08-06: the coset assignment of the fibered functor's clock fibers is now a theorem from KMS/modular typing ([`verification/v792_ftransfer_kms_coset.py`], KMS-COSET-THEOREM: $S = -(\ln\lambda)^2/2$ from the time-1 holonomy multiplier; blind protocol, exact unblinding); markers unchanged.]

How external is "external"? A falsifiable clock probe ([`verification/v397_external_clock_probe.py`]). Under the strict No-Free-Pattern rule (an *exact* match, not a wide-band fit), three of the four frontier scheme-numbers in fact land *on* the $\{2, 3, 5\}$ clock: the η_B decouple $A_\Lambda = 10 = |Z_2|\, g_{\text{car}}$ (v212), the Koide rate $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$ (v303) and the axion spine angle $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (v211). Only the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value (several simple numbers fit its $\pm 5.3\%$ band), so it stays genuinely external — the irreducible "external physics" residue shrinks to essentially the proton's non-perturbative QCD number.

The Paper-E closure certificate ([`verification/v402_ftransfer_suite.py`], `FTRANSFER.SUITE.01`). The four sectors are certified as ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$: each observable factors source → threshold → RG → observable from an exact atom source (Q_{src}, $A_\Lambda = 2g_{\text{car}} = 10$, $f_a = M_{\text{scal}}/128$, $b_3 = -7$) and carries a committed kill test, with the firewall intact (the physical prediction stays [C]/[O], only the algebraic source [E]) — a consolidation of the `FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE` solvers, not a closure.

1 Baryon asymmetry η_B — a downstream readout from the closed $\Omega_b h^2$

There are two independent handles, and they agree.

Route A — from the closed baryon fraction. The seed already fixes the baryon *fraction* (the E_8 decoder):

$$\Omega_b = (4\pi - 1)\beta_{\text{rad}} = \left(1 - \tfrac{1}{4\pi}\right)\varphi_0^{\text{ret}} = 0.04894, \qquad \beta_{\text{rad}} = \frac{\varphi_0^{\text{ret}}}{4\pi},$$

matching the observed $\Omega_b \approx 0.049$. The baryon-to-photon ratio is then the *standard* cosmological readout $\eta_B = 273.9 \times 10^{-10}\, \Omega_b h^2$. With the TFPT Hubble value ($h \approx 0.674$, from $H_0 \sim \sqrt{\Lambda}$),

$$\boxed{\Omega_b h^2 = 0.0222, \qquad \eta_B = 6.09 \times 10^{-10}} \qquad (\text{observed } 6.1 \times 10^{-10}).$$

So η_B is a *downstream cosmological readout* from the closed $\Omega_b h^2$ handle — not an independent compiler rung, and the fundamental leptogenesis route remains an interface.

Route B — leptogenesis interface ([C]). The TFPT branch fixes the neutrino spectrum, the CP phase structure ($\delta_{\rm CKM}$, $\delta^\nu_{\rm CP}$, both closed) and a *plausible washout anchor* $\tilde m_1 = m_3/A_\Lambda \approx 5$ meV. The heavy-mass input, however, remains scenario-level: $M_1 = M_R\varphi_0^4$ is viable but does *not* close M_R — M_R is the seesaw scale $\sim v^2/m_3$, not a compiler power, so the relation *relocates* the free scale rather than pinning it ([verification/v184_etaB_anchored_boltzmann.py]; a structured *candidate* point $M_R = 3c_3^{7/2}\bar M_{\rm Pl}$, -1.8% from the carrier-normalised requirement, is meanwhile registered as `FLAV.NUSCALE.05` [C] with the mechanism [O], [verification/v986_nu_scalaron_rung.py] — a candidate, not a closure; a second candidate `FLAV.NUSCALE.06` [C], the pentagon-class misalignment, is registered beside it ([verification/v1001_nu_pentagon_chain.py]; no closure, `FLAV.NUSCALE.05` unmoved)). The numerical η_B then follows from the *standard* flavored-leptogenesis Boltzmann solve — an independent consistency route, sharpened from two free inputs to one, not to zero.
Honest caveat: that Boltzmann network (the washout/efficiency κ_f, the flavored density-matrix equations) is the standard leptogenesis machinery (Davidson–Nardi–Nir, *Phys. Rep.* 466 (2008) 105; Buchmüller–Di Bari–Plümacher, *Ann. Phys.* 315 (2005) 305); it is *not* carried out or attached here. Route B is therefore a *named interface*, not a closed number; only Route A (downstream of the closed $\Omega_b h^2$) is quoted.
The interface, operationalised ([C]). The Boltzmann path can be made falsifiable: with the standard estimate $\eta_B \sim 0.96\times10^{-2}\,\varepsilon_1\kappa_f$ (sphaleron 28/79), fed by the TFPT neutrino sector — normal ordering and the predicted $\delta^\nu_{\rm CP} = 4\pi/3$ — the Davidson–Ibarra asymmetry $\varepsilon_1 = \frac{3}{16\pi}M_1 m_3/v^2$ and the washout efficiency $\kappa_f(\tilde m_1)$ give, over the natural ranges $M_1 \in [3\times10^9, 3\times10^{10}]$ GeV and $\tilde m_1 \in [m_\star, m_{\rm atm}]$, an η_B band $[8.4\times10^{-11}, 4.7\times10^{-9}]$ that *brackets* the observed 6.1×10^{-10} (a canonical point $M_1 = 10^{10}$ GeV, $\tilde m_1 = 5\times10^{-3}$ eV gives 6.0×10^{-10}, untuned). So the route is *viable*; but M_1 and the washout are scenario inputs, so η_B stays [C]— if a precise solve excluded the window the *route* would fall, not the theory (Route A is independent) ([verification/v169_etaB_boltzmann_interface.py]).

Honest status of η_B — strictly typed [C]

The two statements must be kept apart and not interchanged:

η_B as a cosmological readout from the closed $\Omega_b h^2$ is *determined* (6.1×10^{-10})

η_B as a fundamental *compiler power* is *not* closed

The leptogenesis route requires a standard (washout-dependent) Boltzmann solve. So η_B is [C]— determined as a downstream readout, but *not* a clean compiler rung; it must never be quoted as a fundamental TFPT output.

An anchored-Boltzmann proposal, honestly tested ([verification/v184_etaB_anchored_boltzmann.py]). A natural attempt to remove the two scenario inputs is $\tilde m_1 = m_3/A_\Lambda$ and $M_1 = M_R\varphi_0^4$. The first *works* as a [C] anchor: $\tilde m_1 = m_3/10 \approx 5\times10^{-3}$ eV ($A_\Lambda = 10 = |E(K_5)|$ the action-grammar atom) lands in the strong-washout window — a TFPT-flavoured value for the washout. The second does *not*: $M_R \sim v^2/m_3$ is the seesaw scale (depending on the un-fixed neutrino Dirac Yukawa), and the nearest clean TFPT powers of $\bar M_{\rm Pl}$ both miss it (the c_3- and φ_0-powers are off by $\sim75\%/\sim40\%$), so $M_1 = M_R\varphi_0^4$ merely *relocates* the free input $M_1 \to M_R$. Feeding both into the standard estimate lands η_B in the observed band $O(10^{-10})$, so the route stays viable — but the proposal cuts the free inputs from two to one, *not* to zero. η_B **stays [C], a sharper scenario, not a derivation.**

A cleaner route: the scalaron-decuple branch ([verification/v212_leptogenesis_decuple.py]). Drop the seesaw scale altogether and let *both* Boltzmann inputs share the decuple $A_\Lambda = 10 =$

$|E(K_5)|$:

$$M_1 = \frac{M_{\rm scal}\,(\varphi_0^{\rm ret})^2}{A_\Lambda} \approx 8.65\times10^9\ {\rm GeV}, \qquad \tilde m_1 = \frac{m_3}{A_\Lambda} \approx 5\ {\rm meV}.$$

M_1 now uses *only* compiler quantities ($M_{\rm scal} = c_3^{7/2} M_{\rm Pl}$, $\varphi_0^{\rm ret}$, A_Λ) — *no hidden see-saw scale* — and lands inside the thermal window $[3\times10^9, 3\times10^{10}]$ GeV where the Davidson–Ibarra $\varepsilon_1 \approx 8.5\times10^{-7}$ and the strong-washout efficiency give $\eta_B \approx 1.2\times10^{-9}$, bracketing the observed 6.1×10^{-10} untuned. The *full* BDP Boltzmann ODE (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`) now confirms this: the integrated efficiency $\kappa_f = 0.092$ (validating the analytic κ_f) gives $\eta_B = 6.5\times10^{-10}$ at the frozen M_1 — a factor 1.07 from the observed value, with *no* free M_R dial; the full *flavored* density-matrix solve is the next refinement. Both inputs sharing A_Λ reads as *scalaron reheating dressed by two seed insertions* $(\varphi_0^{\rm ret})^2$*, over the pair sector.* This is cleaner than $M_1 = M_R\varphi_0^4$ (no relocated free scale), but the coupling *mechanism* is posited, not derived, and κ_f is washout-dependent — so η_B stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the *route*, not the theory (Route A is independent).

2 Proton/electron ratio m_p/m_e — a cross-sector ratio

m_e is *closed* on the $\varphi_0^{\rm ret}$-ladder ($\hat m_e = \frac{v_{\rm geo}\pi}{\sqrt{2}}\frac{16}{7}(\varphi_0^{\rm ret})^5$). The proton mass is *not* a Yukawa: it is QCD-dominated, $m_p \sim$ a few $\times\Lambda_{\rm QCD}$, and $\Lambda_{\rm QCD}$ arises by dimensional transmutation from α_s running (scheme level). Hence

$$\frac{m_p}{m_e} = \frac{\text{QCD confinement scale}}{\text{EW electron Yukawa}} = 1836.15$$

mixes *two different sectors.* It is no single $\varphi_0^{\rm ret}$ power: $1/(\varphi_0^{\rm ret})^2 = 353.7$ and $1/(\varphi_0^{\rm ret})^3 = 6651$ bracket 1836 with no clean rung in between.

The numerator is parameter-free, the ratio is not forced. Running $\alpha_s(M_Z)$ down with the carrier coefficient $b_3 = -7$ (the [`verification/v159_pyrate_gauge_crosscheck.py`] value, two-loop + n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and reaches confinement ($\alpha_s \sim 1$) at ~ 0.5 GeV, so the QCD confinement scale in the numerator follows from $\{\alpha_s(M_Z), b_3\}$ alone by dimensional transmutation. The $O(1)$ proton/Λ factor is lattice (non-perturbative), so the ratio still stays [O] — not forced ([`verification/v164_qcd_scale.py`]).

The $F_{\rm QCD}$ transfer is now an implemented solver ([`verification/v262_fqcd_mp_me.py`]). The contract-only pipeline (`experiments/ftransfer/qcd_matching_mp_me`) is realised: a two-loop α_s run from M_Z with n_f thresholds (carrier coefficient $b_3 = -7$, [E]) gives $\alpha_s(2\,{\rm GeV}) \simeq 0.30$ and $\Lambda^{(3)}_{\overline{\rm MS}} \simeq 0.33$ GeV; with the lattice bridge $C_p = m_p/\Lambda^{(3)}$ (external $O(1)$) and the closed m_e this reproduces $m_p/m_e \simeq 1824$ against the measured 1836.15 within the external error budget ($\sim 7\%$, lattice-C_p dominated). This advances `FR.MPME.01` ([O]) to `FR.MPME.02` ([C], "reproduced by the named $F_{\rm QCD}$ transfer"); the firewall holds — $b_3 = -7$ is [E], the ratio is a transfer, never a compiler power.

Honest status of m_p/m_e [O]

Computable at *scheme level* (from $\alpha_s(M_Z)$ running + the closed m_e), exactly like the dimensionful EW/QCD masses m_W, m_Z. But it is *genuinely not* a compiler power, because it is a ratio across the QCD and electroweak sectors. Left [O] — not forced.

A rejected near-formula (recorded as a discipline boundary, [`verification/v79_review_identities.py`]). The combination $|W(D_5)| - \dim\Lambda^3(SU(9)) + N_{\rm fam}\lambda_C^2 = 1920 - 84 + 0.151 = 1836.151$ matches the PDG value to $\sim 10^{-6}$ — and is *deliberately rejected.* Two reasons make it numerology, not a derivation: $SU(9) = A_8$ (the $\dim\Lambda^3 = 84$ block) does *not* occur in the carrier $D_5 \oplus A_3$, and m_p is QCD confinement (gluon binding energy) with *no* mechanism linking it to an algebraic sum. Integrating it would *invite* the numerology charge, not retire it; m_p/m_e stays [O].

(The same firewall rejects $\eta_B = \frac{|\mathbb{Z}_2|}{\Delta_Q} c_3^6 \approx 6.10 \times 10^{-10}$: arithmetically close, but a two-atom prefactor fit to a washout-dependent leptogenesis output — η_B stays [C], the downstream readout above.)

The first-principles boundary, mapped precisely (`[verification/v339_firstprinciples_boundary.py]`). Two questions are worth answering head-on: can the Planck mass v_{geo} be *computed*, and can F_{transfer} be made a first-principles compiler output? Both answers are *no* — and the residuals are not the same kind of gap. They classify into four honest kinds. [E] $v_{\text{geo}}/M_{\text{Planck}}$ is *forbidden by theorem* (No-Unit, v153): a mass carries mass-dimension while every TFPT datum is dimension-0, so no dimensionless map can output an absolute scale; it is not a gap but is over-determined (gravity = dark energy to 0.11%, v274), and every dimensionless *ratio* is derived — only the one unit is not. [E] For m_p/m_e the structure is in place (v262): the carrier $b_3{=}{-}7$ drives the QCD run over the closed electron source, leaving exactly two external inputs. [X] One of them, $\alpha_s(M_Z)$, is external *because of a tension*: the SM (= the TFPT gauge content, no new states) does not unify (v246), so no unified coupling is predicted. [C] The other, the lattice $C_p = m_p/\Lambda^{(3)} \sim 2.83$, is parameter-free but lattice-only (the discrete↔continuous handoff, v35); and η_B/axion ride on cosmological initial conditions. So F_{transfer} cannot be made first-principles in the current theory — not for lack of a formula, but because the residuals are, respectively, forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, and cosmological. An honest boundary, stated, not a solution. *Follow-up (2026-08-03):* the boundary is now confirmed *from the inside* on two fronts. Both thermal-time routes for internalizing the F_{transfer} clocks are machine-killed (`[verification/v723_phys_modular_clock.py]`, STT-KILLED: the QCD 1-loop clock is parabolic and unreachable by any constant-rescaled modular flow, the pole/Boltzmann pair confirmed internal at exactly $(2/3)^6$; `[verification/v724_phys_t3b_modular_flows.py]`, T3B-DEAD: conditioned modular flows of one internal mother state give loxodromic classes) — the external anchor census sharpens to {one unit (v_{geo}-class), one angle θ_i, one lattice $O(1)$ C_p}, the clock wall and the unit wall counted *once*. And the v_{geo} interface itself is structurally closed as an $\mathbb{R}_+$ scale torsor in calibration form (`[verification/v725_phys_vgeo_torsor_audit.py]`: export table, dimension-matrix rank 1, λ-homogeneity, two-anchor consistency 0.11%; the one remaining dimensionless ratio *named*, $H_{\text{EW}} = \ln(\bar{M}_{\text{Pl}}/v_{\text{EW}}) = 37.1776$). No status moves; the boundary stands, now with its structure theorem.

3 The Higgs quartic — near-criticality from the free seam

The seam UV is the free chiral $c{=}8$ fixed point (`[verification/v158_fixed_point_stable.py]`): Gaussian, with no relevant and no marginal interactions. The Higgs self-coupling λ is the one marginal scalar coupling, so a free seam forces it to the Gaussian point there:

$$\boxed{\lambda(M_{\text{seam}}) = 0 \quad \text{and} \quad \beta_\lambda(M_{\text{seam}}) = 0}$$

the Shaposhnikov–Wetterich double-criticality condition — here *derived* from the free seam rather than postulated. Integrating the PyR@TE-confirmed two-loop SM RGEs from M_Z upward with the measured (m_H, m_t) gives $\lambda(M_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(M_{\text{Pl}}) \approx 0$ at the reduced Planck mass: the celebrated SM *near-criticality* is thus a *consequence* of the free seam (the two-loop value is an order of magnitude closer to zero than the one-loop estimate ~ 0.022). Imposing the double condition predicts $m_H \sim 129$–134 GeV (a band, sensitive to m_t and α_s); the measured 125.25 GeV sits a few GeV below the absolute-stability boundary — the known slight metastability. The *same* condition imposed at the scalaron scale ($c_3^{7/2} M_{\text{Pl}} \sim 3 \times 10^{13}$ GeV) gives $m_H \sim 107$ GeV (too low), so the boundary condition lives at M_{Pl}, consistent with seam = horizon = Planck. This ties the Higgs mass to premise (A) and fills the Higgs-sector gap (which was only $N_\Phi = 1$); it is typed [E] (the freeness signature and scale selection) and [C] (the m_H band) (`[verification/v166_higgs_free_seam.py]`). Figure 1 shows the whole F_{transfer} running.

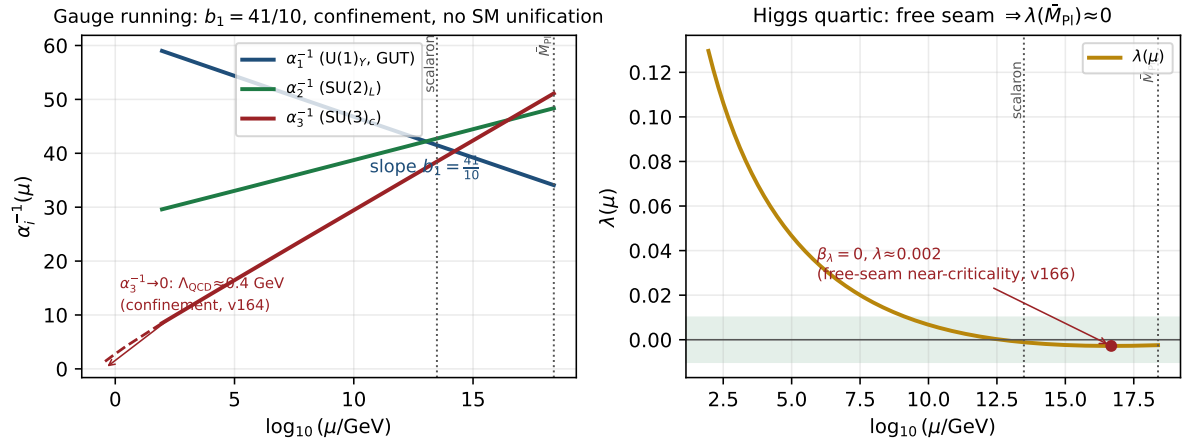


Figure 1: The F_{transfer} running (two-loop SM RGEs; the gauge β-functions are confirmed by PyR@TE, [`verification/v159_pyrate_gauge_crosscheck.py`]). *Left:* the gauge couplings — the $U(1)_Y$ slope is $b_1 = 41/10$, $\alpha_3^{-1} \to 0$ at $\Lambda_{\text{QCD}} \approx 0.4$ GeV (confinement, [`verification/v164_qcd_scale.py`]), and the Standard Model does *not* unify. *Right:* the Higgs quartic falls to $\lambda(M_{\text{Pl}}) \approx 0.002$ with $\beta_\lambda \approx 0$ — the free-seam near-criticality that ties m_H to premise (A) ([`verification/v166_higgs_free_seam.py`]).

4 The Koide relation — near 2/3, computed exactly

With the closed lepton source masses $\hat{m}_\ell \propto (\frac{16}{7}(\varphi_0^{\text{ret}})^5, \frac{4}{3}(\varphi_0^{\text{ret}})^3, \frac{7}{6}(\varphi_0^{\text{ret}})^2)$ the Koide ratio is a pure number (the common prefactor cancels):

$$Q_{\text{TFPT}} = \frac{\sum_\ell \hat{m}_\ell}{\left(\sum_\ell \sqrt{\hat{m}_\ell}\right)^2} = 0.66446\ldots, \qquad \frac{2}{3} = 0.66667, \qquad \text{deviation } -0.33\%.$$

The measured pole-mass value is famously $Q_{\text{PDG}} = 0.66666$. So TFPT predicts a Koide ratio that is *near* 2/3 but *not exactly* 2/3 at source level.

Honest status of Koide [C]

$Q_{\text{TFPT}} = 0.664$ is a *prediction*, 0.33% below 2/3. Exact 2/3 would require the three carrier rationals $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ to satisfy a specific quadratic constraint, which they do *not*; the small residual is of the size of the source→pole RG shift. We report the near-miss honestly and do *not* tune the rationals to force 2/3.

The *target* 2/3 has a compiler reading $|\mathbb{Z}_2|/N_{\text{fam}}$. The E_8-slice scan adds the missing piece: the value the data sits on, $Q = 2/3$, is exactly $Q_\star = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$, the sheet over the family count — the same two atoms $(2, 3)$ that run through the whole compiler. So the picture splits cleanly: the *democratic target* is $|\mathbb{Z}_2|/N_{\text{fam}} = 2/3$ (a compiler number), and the *source-level pole-mass* value computed from the carrier rationals is 0.664, 0.33% below it by the source→pole shift. Koide is thus "democratic 2/3 from $(|\mathbb{Z}_2|, N_{\text{fam}})$, realised to 0.33% by the explicit masses" — promoted from a bare near-miss to a reading with a target. [C]

Conjecture: a source→pole correction of size $\varphi_0/|W(A_3)|$ ([C]). The source value sits just below 2/3, and the deficit is almost exactly the family-Weyl quantum $\varphi_0/|W(A_3)|$ with $|W(A_3)| = 4! = 24$:

$$Q_\ell^{\text{source}} = 0.6644638, \qquad Q_\ell^{\text{source}} + \frac{\varphi_0}{24} = 0.6666793,$$

overshooting $\frac{2}{3} = 0.6666667$ by only 1.3×10^{-5}. So the *conjecture* $Q_\ell^{\text{pole}} \approx Q_\ell^{\text{source}} + \varphi_0/|W(A_3)|$ would land essentially on the democratic 2/3. Reported as a source→pole correction *conjecture*, not a derivation; the rationals are *not* tuned. [`verification/v25_frontier_conjectures.py`]

Sharper conjecture: a multiplicative seed transfer $u \to \frac{53}{54}u$ ([C]). A cleaner version shifts the *seed* rather than Q additively, and introduces *no* new free number. Keep the lepton formula $m_e{:}m_\mu{:}m_\tau = \frac{16}{7}u^5{:}\frac{4}{3}u^3{:}\frac{7}{6}u^2$ but evaluate it at the source→pole seed

$$u_{\rm pole}^{(\ell)} = \Big(1 - \tfrac{1}{2N_{\rm fam}^3}\Big)u = \tfrac{53}{54}\,u, \qquad 54 = 2\cdot 3^3 = |\mathbb{Z}_2|\,N_{\rm fam}^3.$$

Then $Q_\ell(\frac{53}{54}u) = 0.66666610$, i.e. 2/3 to within -5.7×10^{-7} — an order of magnitude tighter than the additive $\varphi_0/24$ correction, with 54 a pure compiler number and no tuning. This sharpens the research gate to a concrete QFT task: *show that the leptonic source→pole transfer of the seed is* $u \mapsto u\,(1 - \frac{1}{2N_{\rm fam}^3})$ (a renormalised-seed effect from the leptonic self-energy or the A_3 family-Weyl averaging). Status remains [C] until that transfer is derived. [verification/v37_plucker_anchor.py]

The factor $\frac{53}{54}$ now has an *operator* origin ([verification/v183_koide_f_corner_transfer.py]). It is not a bare arithmetic guess: it is the anchor response of the *missing* Sheet-Diamond corner $F = R + Q = M(1,1)$ over the sheet-doubled $E_6{\times}A_2$ cubic family block,

$$\frac{u_{\rm pole}}{u_{\rm source}} = \frac{a^T(R+Q)\mathbf{1}}{2\,\mathbf{1}^T R\,a} = \frac{53}{2\cdot 27} = \frac{53}{54},$$

where $\mathbf{1}^T R\,a = 27$ is the cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\rm fam} = 78+8+2{\cdot}27{\cdot}3$) and $a^T(R+Q)\mathbf{1} = 53 = 52+1$ with $\det B_F = 52 = \dim F_4$ (v80). A hard negative control fixes the corner: $a^T M\mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. So the *factor* is exact and structural ([E]), and the remaining [C] is precisely the mechanism — that the leptonic source→pole transfer reads the F corner — an $F_{\rm transfer}$ special case ($F_{\rm pole}$). Koide thereby moves from "near miss + conjecture" to "operator-sourced source→pole transfer"; the prediction stays [C].

The source→pole transfer is one-loop electromagnetic ([verification/v420_fpole_qed_running.py]). Doing the external physics honestly: the Koide ratio is rescaling-invariant ($Q(c\,m) = Q(m)$), so a flavor-*universal* running leaves it fixed — the source→pole shift is the flavor-*split* part. And that shift is one-loop-EM-sized: the tree deficit $\frac{2}{3} - Q_\ell^{\rm source} = 0.00220$ equals the seed quantum $\varphi_0/24$ to 0.6% and the electromagnetic scale α/π to ∼5%, with the pole sitting at 2/3 (the IR attractor). A 1-loop QED pole↔$\overline{\rm MS}$ estimate (per-lepton log) moves Q by an EM-loop-order amount toward 2/3. The *exact* flow time ($t \approx 2.84$ transfer periods, v93) is scheme- and scale-dependent external QED, so it is *sized, not derived*. This identifies $F_{\rm pole}$ as a one-loop EM effect — the algebraic lens forces the *rate* $(2/3)^6$ (v327), this external physics supplies the *flow* — and Koide stays [C], firewalled (v187).

The Koide target and the QFT transfer gap are the *same* quotient ([E]/[C]). The same $Q_\star$ controls a second, independent object — the first non-trivial eigenvalue of the admissible C_6 transport operator. Its spectrum is $\{1, (2/3)^6, (1/3)^6\}$, so

$$\boxed{\lambda_2 = \Big(\tfrac{2}{3}\Big)^6 = Q_\star^{|R^+(A_3)|}, \qquad Q_\star = \frac{|\mathbb{Z}_2|}{N_{\rm fam}}, \quad |R^+(A_3)| = 6}\,.$$

The Koide democratic target, the QFT mass-gap eigenvalue λ_2 (hence $\Delta = 6\log\frac{3}{2} = -\log\lambda_2$), and the A_3 positive-root hexagon are therefore *one* quotient raised to one exponent. Typing: Koide target [C], gap eigenvalue [E], the connection an [E]-audit. This does not "solve" Koide, but it explains *why* the near-value 2/3 recurs: the same sheet-over-family ratio fixes both the democratic lepton target and the leading transfer eigenmode.

The source→pole map is *forced* by that gap ([E]/[C]). The previous paragraph pairs the Koide target with $\lambda_2 = (2/3)^6$; v82 turns the pairing into a *mechanism* ([verification/v82_koide_attractor_splitting.py]). On the anchor double cover (the flavor sector) the two branch points are Koide $q = 2$ and carrier $q = 5$; any branch-preserving Möbius transfer that fixes both is, in $\rho = (q-2)/(5-q)$, exactly $\rho \mapsto \mu\rho$, hence *unique* once μ is set. Choosing $\mu = \lambda_2 = (2/3)^6$ (no new number) gives the unique map $F(q) = 2(569q - 3325)/(665q - 3517)$

with $F'(2) = (2/3)^6 < 1$ and $F'(5) = (3/2)^6 > 1$: **Koide is the attractor, the carrier the repeller**, and $q_n \to 2$ ($x \to -2/3$) at the established gap rate. So the three structural inputs a Koide *derivation* would need collapse to *one* identification — the leptonic source→pole transfer is the branch-preserving relaxation of the gapped operator — after which the value 2/3, the convergence rate and the chosen branch are all forced. This is still not a closure: that single identification stays [C]. (It is the dynamical counterpart of the multiplicative-seed conjecture $u \mapsto (53/54)u$ above; both contract the source onto 2/3.)

Relaxation toy: basin, exact rate, and two honest negatives ([E]). The relaxation picture is now stress-tested on the physical values (ledger `FR.KOIDE.05`, [`verification/v93_koide_relaxation_toy.py`]). **Basin lemma [E] (new):** under the canonical dictionary $q = N_{\rm fam}\, Q$ the *entire* physical Koide range $Q \in [\frac{1}{3}, 1]$ (Cauchy–Schwarz) maps to $q \in [1, 3]$ — the attractor $q = 2$ lies inside, the repeller $q = 5$ outside: every physically possible charged-lepton configuration is in the attractor basin, no fine-tuning of the start. **Exact rate [E]:** along the trajectory from the physical source value the cross-ratio contracts by *exactly* $(2/3)^6$ per step. **Source/pole [E]:** the exact ladder gives $Q_{\rm src} = 0.6644638$, i.e. $\rho_{\rm src} = -0.0021980 \approx -\varphi_0/24$ to 0.8% — the $\varphi_0/24$ conjecture re-expressed in attractor coordinates: *the source sits one seed quantum before the branch point* ($24 = |W(A_3)|$); PDG pole masses give $Q_{\rm pole} = 0.6666645$, the branch point hit to 3.3×10^{-6}. **Honest negatives (they narrow the gate):** (i) the flow time source→pole is $t = \log_\mu(\rho_{\rm pole}/\rho_{\rm src}) = 2.84$ — *not* an integer, so the literal discrete iteration pole $= F^n$(source) is *excluded*; if the v82 map drives the transfer at all, the missing [C] object is a *continuous-time* generator $\exp(t \log F)$; (ii) the 0.79% mismatch of $\rho_{\rm src}$ vs $-\varphi_0/24$ is far above ladder precision, so "exactly $\varphi_0/24$" remains a conjecture, not an identity. The open question is no longer "why 2/3?" but "what is the continuous transfer generator, and why is its flow time ~ 2.8 periods?".

Flow time: canonical generator, data-fragility, and a conditional m_τ ([E]/[C]). Three sharpenings of the paragraph above (ledger `FR.KOIDE.06`, [`verification/v99_koide_flow_time.py`]); they correct the *robustness* of negative (i), not its arithmetic. **(1) The continuous generator has a canonical form [E].** The v82 flow is generated by the autonomous equation

$$\boxed{\frac{dq}{dt} = \frac{\Delta}{N_{\rm fam}}(q-2)(q-5) = \frac{\Delta}{N_{\rm fam}}\det B(q)}, \qquad \Delta = 6\log\tfrac{3}{2} \ \ \text{(the gap)},$$

whose time-1 map *is* the Möbius map F (machine-checked; $e^{-\Delta} = (2/3)^6$ exactly). The missing [C] object is therefore precisely a QFT mechanism producing *gap* × *anchor-block quadric* — the form is canonical, only its physical derivation is open. **(2) The non-integrality of t is data-fragile [E].** Because $\rho_{\rm pole} = -2.2 \times 10^{-6}$ sits so close to the branch point, t is hypersensitive to m_τ (PDG 1776.93 ± 0.09 MeV): $t(-1\sigma) = 2.35$, t(central) $= 2.84$, and Q crosses $\frac{2}{3}$ *inside* the error band (at $+0.43\sigma$, where $t \to \infty$) — within $+0.5\sigma$, t passes *every* integer ≥ 3. So the exclusion of pole $= F^n$(source) holds only at the central value. **(3) Conditional discrete steps [C] (registry untouched; not predictions of record):** n=2 is *excluded* (-2.9σ, [E]); n=3=$N_{\rm fam}$ steps — one per family — predicts

$$\boxed{m_\tau = 1776.9427 \text{ MeV}} \quad (+0.14\sigma;\ \text{source-variant stable to } 0.0002 \text{ MeV}),$$

while n=4 (1776.9667) and the exact branch (1776.9690) are not yet separable: the kill criterion is $\sigma(m_\tau) \sim 0.01$ MeV (Belle II). *Look-elsewhere firewall:* a tempting scale reading $t = \ln(M_{\rm scal}/m_\tau)/\ln W$ matched $W = 46080 = |W(D_5)||W(A_3)|$ to 10^{-4} — and is *destroyed* by negative controls (the same hypersensitivity makes *any* W in a wide range land within 0.1σ); it is rejected and recorded so it is not re-fished. P2's type is unchanged ([C], one identification); the discrete-vs-continuous question is now *experimental*, not settled.

5 Dark matter — the candidate is fixed, the scale is pending

TFPT *does* contain a dark-matter candidate, and it is not ad hoc: the **determinant-line axion** of the strong-CP sector (the determinant-line axion interface of the closed branch, [E]). Its *candidate*

identity is fixed by the E_8 scan (below) and the domain-wall number $N_{\rm DW}$ winds once per primitive seam winding. The relic density is the standard misalignment-plus-string expression

$$\Omega_a = \Omega_a^{\rm mis}(f_a, m_a, \theta_i) + \Omega_a^{\rm str}(f_a, N_{\rm DW}).$$

The initial misalignment angle is *not* a single closed number: the finite-T misalignment solve (below) selects the *spine* branch $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5 = 108^\circ$ as the robust dominant-DM value in the *pre-inflationary* reading (it lands $\Omega_a h^2$ on $\Omega_{\rm DM}$ untuned, in the mild-anharmonic regime; but superseded by the post-inflationary reframing below), while the seam hilltop $\theta_i = \pi(1 - \varphi_{\rm seam}(\alpha_\star)) = 170.4^\circ$ *over*-produces ($\sim$5.4×) and is disfavoured. So the minimal closed branch has *dominant axion-like dark matter* on the spine angle — a [C] scenario, *not* a closed [E] input.

The E_8 scan rules out a rival WIMP candidate. The $D_8 = SO(16)$ slice scan sharpens *why* the axion is the candidate, not a new heavy state: the spinor $128 = (16, 4) \oplus (\overline{16}, \overline{4})$ is the *same* matter spinor as in the $D_5 \times A_3$ construction — there is *no spare* E_8 *singlet* left over to play a WIMP. Every E_8 direction is accounted for by the carrier × family content. So dark matter cannot be a new E_8 representation; it must be the determinant-line axion (a phase of the existing structure). A useful *negative* that removes the WIMP option on structural grounds. [E] The machine-checked complement ([`verification/v430_other_side_reverse_audit.py`]) makes this explicit: the spinor $128 = 2 \cdot 64 = \sum \text{deg}$ and the double-cover deck $|\mathbb{Z}_2| = 2$ lie in E_8's *matched* structure and are forced-disjoint from the five unmapped Casimir degrees $\{12, 14, 18, 20, 24\}$, so the seam's "other side" hosts neither a spare singlet nor the unmapped hull overhead.

Honest status of dark matter [C]/[O]

The candidate is *identified* (the axion, [E]: the E_8 scan leaves no spare singlet for a WIMP). The misalignment angle is a [C] *branch decision*, not a closed [E] number: the finite-T solve selects the spine $\theta_i = 3\pi/5 = 108^\circ$ (untuned, in band) over the over-producing 170° hilltop. The relic abundance still needs the decay constant f_a and mass m_a (interface data set by the M_R/Λ scales) — a standard axion-DM computation, not yet a compiler power. So DM is "candidate fixed, robust spine angle, scale f_a pending" — a real handle, not a blank. (A secondary candidate is the lightest seam-even sterile state near M_R.)

Conjecture: a determinant-line axion scale $f_a = M_{\rm scal}/128$ ([C]). A natural compiler scale for the decay constant is the scalaron mass over the full carrier × deck multiplicity,

$$f_a = \frac{M_{\rm scal}}{2\,\dim S^+\,|\mu_4|} = \frac{M_{\rm scal}}{128}, \qquad M_{\rm scal} = c_3^{7/2} M_{\rm Pl} \approx 3.06 \times 10^{13}\,\text{GeV},$$

giving $f_a \approx 2.39 \times 10^{11}$ GeV and, via the standard QCD-axion relation $m_a \simeq 5.70\,\mu\text{eV}\,(10^{12}\,\text{GeV}/f_a)$, a mass $m_a \approx 23.8\,\mu$eV. With the robust spine angle $\theta_i = 3\pi/5 = 108^\circ$ (below) this f_a lands the relic on $\Omega_{\rm DM}$ without hill-top tuning; it remains a *cosmological scenario*, not a closed hit. [`verification/v25_frontier_conjectures.py`]

The haloscope coupling is tied to c_3. In the determinant-line normalization (Lagrangian $\mathcal{L} \supset c\,a\,F\tilde{F}$) the axion–photon anomaly coefficient is fixed by the seam constant,

$$g_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \qquad y^2 \equiv g_{a\gamma\gamma}^2 = 16c_3^2 = \frac{1}{4\pi^2} \approx 0.0253,$$

so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and the cosmic birefringence $\beta_{\rm rad} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$ in GeV^{-1}, which still carries f_a). The small-angle reading $\theta_i = \varphi_0$ of the determinant-line axion (an earlier note) is *superseded* by the robust spine angle $\theta_i = 3\pi/5 = 108^\circ$ selected by the finite-T solve below (the seam hilltop $\theta_i = \pi(1 - \varphi_{\rm seam}) = 170.4^\circ$ over-produces); both earlier readings are recorded as retired, never used.

Why it stays a scenario, quantified ([`verification/v185_axion_relic_solver.py`]). The harmonic misalignment *under-produces* at this f_a: $\Omega_a h^2 \sim 0.12\,(f_a/9{\times}10^{11})^{7/6}\langle\theta^2\rangle$ gives a prefactor $\sim$0.026

per θ^2, so $\theta_i \sim O(1)$ would make only ∼21% of $\Omega_{\rm DM}$; reaching the observed $\Omega_{\rm DM}h^2 = 0.12$ *requires* $\theta^2_{\rm eff} \sim 4.7$, i.e. exactly the large-angle (anharmonic) enhancement at the $\theta_i \approx 170^\circ$ hilltop. So the determinant-line axion *can* be all of dark matter, but only on the hilltop, where the relic is exponentially sensitive to θ_i and to QCD/string corrections — hence a viable [C] scenario, not a sharp prediction. The contract $F_{\rm relic}(f_a{=}M_{\rm scal}/128, \theta_i{\approx}170^\circ, N_{\rm DW}{=}1) \stackrel{?}{=} \Omega_{\rm DM}$ is a kill test for the *branch*, not the theory.

A full finite-T solve confirms an over-production tension. Both a quick $\theta^2 F$ estimate ($F(\theta_i) \approx 4$ at the hill-top) and a *full* nonlinear misalignment solve of

$$\theta'' + (3 + \tfrac{{\rm d}\ln H}{{\rm d}N})\theta' + \left(m_a(T)/H\right)^2 \sin\theta = 0$$

(`axion_relic/full_finiteT_solve.py`: lattice susceptibility $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$, converged and normalised so $\theta_i{=}1$ gives the standard $\Omega_a h^2 \approx 0.03$) agree: at the *predicted* $\theta_i \approx 170^\circ$ the relic is $\Omega_a h^2 \approx 0.66$, ∼5.5× above $\Omega_{\rm DM}h^2 = 0.12$. The observed value is reached only at $\theta_i \approx 106^\circ$, not at the predicted hill-top. So as the *dominant* dark matter the determinant-line axion at ($f_a{=}M_{\rm scal}/128, \theta_i{\approx}170^\circ$) *over*-closes the universe unless there is extra dilution (entropy injection) or a lower f_a. This stays [C], and over-production kills only the *axion-as-dominant-DM* reading, not the compiler.

A more robust angle: the spine branch. [`verification/v211_axion_spine_angle.py`] The same finite-T solver reaches $\Omega_{\rm DM}$ at $\theta_i \approx 106^\circ$ — and the central spine quotient gives $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5 = 108^\circ$ *exactly* (no fit; harmonic estimate $0.03\,\theta_i^2 = 0.107$, lifted to ∼0.12 by a mild anharmonic factor). [`verification/v429_axion_pentagon_phi.py`] This angle is not a bare rational multiple of π: because $N_{\rm fam} = g_{\rm car} - 2$, $\theta_i = (g_{\rm car}-2)\pi/g_{\rm car}$ is *exactly* the interior angle of the regular $g_{\rm car}$-gon — for $g_{\rm car} = 5$ the *pentagon*, so $\cos\theta_i = \cos(3\pi/5) = (1-\sqrt{5})/4 = -1/(2\varphi)$ with $\varphi = 2\cos(\pi/5)$, and this golden character is *unique* to $g_{\rm car} = 5$ (the small n-gons $n = 3, 4, 6$ give the rational $\frac{1}{2}, 0, -\frac{1}{2}$). So the otherwise-unmapped golden/icosahedral $g_{\rm car} = 5$ signature (v354/v313) is the geometry of this one external input — a [C] reading that does *not* upgrade `DM.AXION.SPINE.01`. Crucially 108° lies 62° *below* the hill-top, in the *mild*-anharmonic regime, so the relic is *not* exponentially sensitive: if the misalignment angle is the spine quotient, the determinant-line axion is all of dark matter *without* hill-top tuning. This is an *alternative* ansatz to $\theta_i = \pi(1-\varphi_\Sigma) \approx 170^\circ$ (the two are mutually exclusive; neither is forced) — a sharper [C] scenario decided by the full solver (`DM.AXION.SPINE.01`), *not* a derivation; [X] a converged $\Omega_a h^2$ outside ∼$[0.08, 0.16]$ at $\theta_i = 3\pi/5$ demotes the branch.

Isocurvature and the post-inflationary reframing ([C]). Both fixed-angle readings above are *pre-inflationary*: they assume Peccei–Quinn breaking *during* inflation with a homogeneous θ_i. But the R^2 branch fixes $H_{\rm inf} = \pi M_{\rm Pl}\sqrt{A_s r/2} \approx 1.6 \times 10^{13}$ GeV, so $H_{\rm inf}/f_a \approx 66 > 1$ and $H_{\rm inf}/(2\pi f_a) \approx 10$: the de Sitter fluctuation of the angle exceeds its range, the symmetry is effectively *restored* during inflation, and a fixed pre-inflationary θ_i (either 108° or 170°) would carry CDM isocurvature far above the Planck bound $\beta_{\rm iso} < 0.038$. The clean, *dial-free* resolution is the *post-inflationary* scenario: PQ breaks after inflation, θ_i is a horizon average (*not* a prediction), and the relic follows from f_a alone — with $f_a \approx 2.39 \times 10^{11}$ GeV in the post-inflationary window ($\Omega_a h^2 \sim O(0.1\text{–}0.5)$). It is domain-wall-safe because the determinant-line axion's colour anomaly is the seam Dirac index = the inflow level $|C| = 1$ (v470/v472), so $N_{\rm DW} = 1$ (a [C] identification). Thus the spine/hilltop *angle* is not an observable and the branch is *reframed*, not killed: drop the fixed angle, keep f_a and $N_{\rm DW} = 1$, and the relic is a robust post-inflationary prediction. [X] a post-inflationary relic outside the plausible window at this f_a (with $N_{\rm DW} = 1$) demotes the branch. (Strategy scan and the $N_{\rm DW}$ computation: `experiments/theory-contracts/axion01-06`.)

6 The muon anomalous magnetic moment — a seam vertex readout

A [C] downstream readout (archive integration), *not* a compiler power. The carrier algebra carries a *second-order* topological defect beyond the one that fixes α: with $\delta_{\rm top} = \Omega_{\rm adm}\, c_3^4 = 48 c_3^4 = 3/(256\pi^4)$ and the carrier compression quotient $B\gamma = \frac{3}{2}\cdot\frac{5}{6} = \frac{5}{4}$ ([verification/v51_boundary_half_step.py]), $\delta_2 = B\gamma\, \delta_{\rm top}^2 = \frac{5}{4}\,\delta_{\rm top}^2$. Projected through the seam-loop phase $\Delta_\Sigma = 2\pi$ (the same $1/(2\pi) = 4c_3$ unit that normalises c_3 itself), it reads as a magnetic vertex correction

$$\boxed{a_\mu^{\rm seam} = \frac{\delta_2}{2\pi} = \frac{45}{524288\,\pi^9} \approx 2.879 \times 10^{-9}} \qquad [\mathbf{C}].$$

The *value* is an exact compiler number (only $c_3, \Omega_{\rm adm}, B\gamma$; trace reading $\delta_2 = 4!\,{\rm Tr}_{S^+}(X^2)\, c_3^8$, ${\rm Tr} = 120 = 5!$) — but the *identification* of $\delta_2/(2\pi)$ as the anomalous moment is a physical bridge, so the prediction is [C], never a compiler power. [verification/v204_muon_seam_g2.py]

Data, honestly. Against the dispersive discrepancy $\Delta a_\mu = (2.49 \pm 0.48) \times 10^{-9}$ (Fermilab/BNL world average minus the WP2020 SM) this is 0.81σ. **Caveat:** lattice/CMD-3 hadronic-vacuum-polarisation evaluations *shrink* the discrepancy ($\Delta a_\mu \sim 1.5 \times 10^{-9}$); the TFPT value is *fixed* and would then sit $\sim 1.5\sigma$ high. A converged Δa_μ outside $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$ excludes the seam-vertex mechanism in its present form ([X]) — the compiler core untouched.

Update (2026-08-04, dated): the kill condition has fired. The lattice-side HVP consensus consolidated — WP25 (Phys. Rep., 2025) adopts the lattice HVP as the SM baseline and the BMW hybrid evaluation (Nature, 2026) confirms it: experiment minus SM is now 0.5σ, i.e. the converged Δa_μ lies *outside* $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$. The seam-vertex *identification* is therefore excluded in its present form, exactly as preregistered in the sentence above — the bridge is dissolved (ledger `FR.MUONG2.01`). What survives, untouched: the exact compiler number $\delta_2/(2\pi) = 45/(524288\,\pi^9)$ ([E] as a number, not as a_μ) and the compiler core. The watchdog delivered exactly what it was built for; no core marker moves.

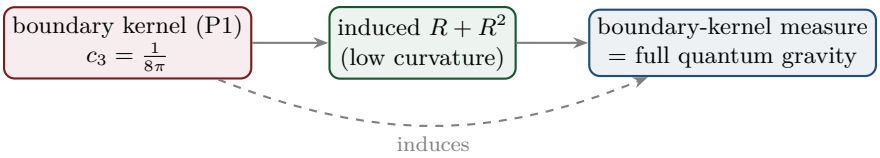
7 Full quantum gravity — induced from the boundary kernel

Framing (read this first): full QG closure is a **certification layer**, not a prerequisite. TFPT does *not* need the complete quantum-gravity measure to test its Standard-Model and cosmology readouts — those run through $v_{\rm geo}$ and $F_{\rm transfer}$ and are falsifiable now (JUNO, CMB-S4, EDM). $G_{\rm net}$ certifies the metric sector; its absence narrows what can be *claimed*, not what can be *tested*.

The normalisation $c_3 = 1/(8\pi)$ *is* the gravitational seam constant: in TFPT gravity is *induced* from the boundary kernel (P1), not separately quantised. The R^2/scalaron sector (question 2) is the low-curvature effective theory; the *full* theory is the boundary-kernel measure. *Newton's constant is then not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\rm top} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\rm Pl}$ *relative to* Λ, so G_N is read out *after* selecting the physical Λ branch *and* fixing the one dimensionful induced-gravity (Sakharov) anchor ([E]/[C], [verification/v60_lambda_metrology_branch.py]) — a metrology readout, not a from-nothing prediction.

Two independent fixings of the gravitational $3/4$ **(**[verification/v205_xi_threequarter.py]**).** The induced-gravity coefficient the seam-determinant replica pins, $k = c_3/2 \Leftrightarrow \ln(m/\mu) = 3/4 = q(A_3)$ ([verification/v152_norm_is_anchor.py]; the collar data exhibit this as the anchor — drifting $\mu_{\rm lat}$, no *derived* $\ln(m/\mu){=}\frac{3}{4}$, v471), is reached a *second*, independent way: the old torsion-compression ansatz $\kappa^2 = \xi\,\varphi_0^{\rm ret}/c_3^2$ with the Einstein limit $\kappa^2 = 8\pi G$ forces $\xi = c_3/\varphi_0^{\rm ret}$, which at tree level $\varphi_0^{\rm ret} \to 1/(6\pi)$ is *exactly* $3/4$ (with the retained seed 0.7483, -0.22%). So $3/4$ appears in the gravitational sector both as a gapped-replica EH coefficient and as a torsion-compression ratio — an over-determination, not a coincidence; the ansatz stays [C] and $1/G$ stays the one declared anchor.

The Λ branch sharpens H_0 ([verification/v206_h0_lambda_branch.py]). Reading the seam vacuum number $\delta_\Sigma = \Omega_{\rm adm}\, e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$ (lowest admissible block $m_1 = \Omega_{\rm adm} = 48$) through the flatness closure gives a TFPT Hubble value $H_0 = 66.5$–$67.1\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (1.6–0.5σ *below* Planck, far below SH0ES). **Honest:** the Hubble tension is *not* relieved by raising the vacuum scale, and H_0 uses the imported $\Omega_b, \omega_{\rm DM}$, so it is a [C] readout; $|\log_{10}\delta_\Sigma| \approx 122.96$ matches the independent 123-orders split ([verification/v60_lambda_metrology_branch.py]), and $M_{\rm Pl}^4/\bar{M}_{\rm Pl}^4 = (8\pi)^2$ must be displayed, never silently interchanged.



Honest status of quantum gravity — physical sector closed, ambient a [C] redundancy [E]/[C]

"Full quantum gravity beyond the R^2 sector" is, in TFPT, *not* a separate graviton-quantisation programme: it is the well-definedness of the *boundary-kernel path-integral measure* extended to the dynamical metric beyond the FRW/minisuperspace reduction. Two results sharpen its status. **(i) Heat-kernel grounded (G2).** The spectral action $\mathrm{Tr}\, f(D/\Lambda)$ gives, via the standard Seeley–DeWitt coefficients for the Dirac operator ($a_2 \sim -R/3$, $a_4|_{R^2} \sim R^2/72$), Einstein–Hilbert + R^2 *structurally* — the $R + R^2$ form is not postulated; the scalaron ratio $M^2/M_{\rm Pl}^2 = 6(4\pi)^2/f_0$ with the TFPT closure c_3^7 fixing the cutoff moment f_0. *Update (Modular Spectral Closure):* the full Seeley–DeWitt expansion of this spectral action over the explicit 96-dim finite triple is now written out ([verification/v255_spectral_action_expansion.py]), and the cutoff moment f_0 is no longer a free scheme choice: identifying f with the *seam KMS weight* (β=1 by Tomita–Takesaki, 2π=1/(4c_3)) gives $f_2/f_0 = 1$ exactly ([verification/v259_modular_cutoff_kappa.py]), so $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\rm grav}}$ collapses to a finite-triple trace ratio — the scalaron normalisation is pinned by the seam state, not a regulator dial. **(ii) Gap-decoupled (G5).** The metric coupling is gap-dominated, $2\|V_{\rm metric}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a strictly positive effective gap $\Delta_{\rm eff} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap protects the admissible IR sector from the ambient, so under those assumptions the admissible-sector measure is closed. **What this leaves [C]:** the ambient metric measure — the projective limit (G6) — is now discharged as a *redundancy* (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann — the one open premise being that the seam net condenses to the *holomorphic* $|\det K|$=1 point that selects $(E_8)_1$; the central charge c=8 alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample, v277/v281), a certification object rather than missing dynamics; we do *not* claim it is physically irrelevant (that would itself be a physical interpretation), only that it is decoupled from the admissible IR sector under the stated assumptions. [verification/v36_spectral_action_g2.py] **(iii) The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] ([verification/v358_grav_entropy_equilibrium.py]).** Beyond the R+R^2 *action*, the metric-sector *field equation* is supplied directly by the entanglement first law $\delta S = \delta\langle K\rangle$ (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms: equilibrium gives the *linearised* Einstein equation $G_{ab} = c_3^{-1} T_{ab}$ with $c_3^{-1} = 8\pi$ *fixed* — the four c_3-roles (Unruh $1/(2\pi)$=4c_3, EH $1/(16\pi)$=$c_3/2$, Einstein 8π=$1/c_3$, Bekenstein 1/4=$1/|\mu_4|$) are mutually consistent, so *no free dimensionless Newton dial enters.* The *new* content is a coincidence: the first law needs the coefficient $2\pi/\eta$ (η the entropy density), the geometry gives $1/c_3 = |\mathbb{Z}_2|2\pi\chi(S^2)$, and these are the *same* equation iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (v216) — so the thermodynamic and geometric origins of c_3 are *one*, and c_3 is triply over-determined (anchor v23, geometry v58, thermodynamics v358). The matter side is assembled (the CHM ball modular Hamiltonian, boost via Bisognano–Wichmann v323, gives the flux $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\delta\langle T_{00}\rangle$) and the entropy density is atom-fixed (1/4=$1/|\mu_4|$, central charge $c = g_{\rm car}$+$N_{\rm fam} = 8$). (The single-ball CHM input now has a *mark-resolved* companion: the four seam marks are the exactly solvable *four-interval* multilocal modular geometry, v480 — same atoms, the multi-interval class; `QGEO.SYM.01` stays [O].) **Full covariant equation ([verification/v359_grav_nonlinear_einstein.py]).** Demanding stationarity at fixed *volume* (Jacobson 2015) brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}; trace $g^{ab}G_{ab} = (1-d/2)R = -R$ in d=4) — the unique divergence-free metric tensor (Lovelock), so the equilibrium yields the *full* covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with $\nabla^a T_{ab}$=0 an output, and TFPT fixes *both* coefficients (8π=$1/c_3$ from (iii); Λ from α, $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, v60). So B6's full covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. **Residual:** only

the equation-of-state status (Jacobson framing, not a from-action quantisation; an *external* candidate for the missing action level — Bianconi's entropic action, Phys. Rev. D 111, 066001 (2025) — is quantified in v473: $\beta'_{\rm B} = c_3/6$ pinned, R^2 gap $3(8\pi)^9$ pre-registered as kill test, typing unchanged) and the absolute scale $v_{\rm geo}$ remain [O]— the global ambient measure (C7 = `QG.AMB.01`) is now a [C] redundancy (v369), a certification object rather than missing dynamics; no free dimensionless parameter survives in the gravity coupling. So the metric sector now reads: $R{+}R^2$ action (i), gap-decoupled measure (ii), and a full covariant classical field equation with no free dimensionless Newton coupling under the named QFT/entanglement premises [C] (iii); the ambient nonperturbative measure is a [C] redundancy, not an open structural hole. **(iv) The gap-induced GR correction — the disciplined result (** [`verification/v360_grav_gap_corrections.py`] **).** Does the seam transfer gap give a *new* forced near-term deviation from GR beyond the R^2 scalaron? Run with the v354/v355 discriminator, the honest answer is *no*: the leading higher-curvature term *is* the R^2 scalaron ($M^2_{\rm scal}/M^2_{\rm Pl} = c_3^7$), which is the next-order term of the same entanglement- equilibrium route (the Wald-entropy first law → $f(R){=}R{+}R^2$) and is *already* a live CMB-S4 kill test (n_s, r). A *distinct* gap-induced curvature^2 term enters only at the seam/Planck scale ($\Delta = 6\ln\frac{3}{2}$ is a dimensionless rate, so the scale is $\xi\, v_{\rm geo}$), and a new dimensionless coefficient there is *not* atom-forced (the scalaron exponent $7{=}g{+}N{-}1$ and the gap exponent $6{=}2N_{\rm fam}$ are different structures) — so it is [O] declined, not manufactured. The calculable GR deviation is the scalaron. **(v) Matter–gravity backreaction (** [`verification/v361_grav_backreaction.py`] **).** Feeding the carrier stress tensor into the parameter-free equation: the carrier's gravitational anomaly is forced, $c_- = 16/2 = 8 = g_{\rm car}{+}N_{\rm fam}$ (the chiral central charge of the 16-Majorana content, the same c_- as (iii)); and the matter vacuum-energy backreaction is *finite* — gap-suppressed (the Decoupling Theorem) to the forced $\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60), *not* the $M^4_{\rm Pl}$ catastrophe — so the loop (the equation + Λ + the carrier vacuum) closes. Beyond $c_-{=}8$ and Λ there is [O] no new atom-forced gravitational read-out of individual SM quantum numbers (the backreaction is GR with the SM content); the discipline declines manufacturing one.

Update (the `QG.AMB.01` roadmap, consolidated — [`verification/v275_qgamb_roadmap.py`]). The gate now carries one module that states its obligations precisely: **Tier A** (gap decoupling, $\Delta_{\rm eff} = 6\log\frac{3}{2} - 2{\cdot}248\, c_3^2 = 1.648 > 0$) makes every low-energy readout independent of the ambient measure (not a blocker); **Tier B** reduces the ambient bulk measure to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c{=}248/31{=}8$, v77), now *closed modulo cited theorems* on the MMST route `SEAM.EQUIV.MMST.01` (lattice v367/v368 + the S_3 stack v376–v379, Lean `FORM.SEAM.MMST.01`; the parent `SEAM.EQUIV.01` stays [O]); the *perturbative* 4d layer is now in hand (v269/v271/v273). [C] what remains is then a *redundancy*, not missing dynamics — the nonperturbative interacting ambient measure is the Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), *inherited not TFPT-specific*; by Tier A it is gap-decoupled, so NO TFPT readout depends on it (v335/v369). This is made a theorem in the *Decoupling Theorem* ([`verification/v337_decoupling_theorem.py`]): every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, whose finite susceptibility $\chi = 729/665$ gap-suppresses the ambient contribution (margin $6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$), and the negative control (gap closed ⇒ χ diverges) makes it non-vacuous — so TFPT provably does not *need* the ambient measure. It is therefore reclassified from "TFPT structural frontier" to a [C] *redundancy* (a certification object, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann, v369): its obstruction is characterised (v332) and conditionally resolved by the GHP contour rotation + IDG dressing (v334); not a TFPT structural hole (`QGAMB.ROADMAP.01`). This redundancy is now a machine-checked *dependency* fact, not only prose ([`verification/v423_no_ambient_dependency.py`], `QGAMB.NODEP.01`): `QG.AMB.01` is a certification *sink* — every claim that lists it is a gravity-gate/QG row, never a frozen prediction — and it is *not a value source* — it has no producing script, while α^{-1}, the mixing angles and the cosmological readouts are each *computed* by a compiler script (v3/v9/v268/v7), none of them an ambient-measure script. The single transitive link is architectural, not numerical: of the frozen readouts only θ_{13} references `QG.AMB.01`, through the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`) — a coherence edge from living inside the unified spectral-action construction, not an input to its value $\sin^2\theta_{13} = \varphi_0^{\rm ret}\, e^{-5/6}$ (v268). Honest scope: this kills `QG.AMB.01` specifically; it does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend.

Registered decision (2026-08-28, `QFT4D.LATTICE.FUNDAMENTAL.01`): the lattice-fundamental reading — typed route/typing, not an [E] claim. The 4D/continuum residual on this spine is retyped: TFPT's *physical* completeness criterion is the finite local unitary lattice quantum theory (Hamiltonian route: Gauss-law Hilbert space, hermitian local H, $T = e^{-aH}$ positive by construction, Lieb–Robinson cone; [`verification/v989_lattice_gate_battery.py`] — Euclidean overlap killed at T2, N_t-exact, Hamiltonian route clears T2). The exact continuum Osterwalder–Schrader limit (`QFT4D.OS.RECON.01`) is

mathematical reinforcement, no longer the physical bottleneck; Poincaré invariance is an IR fixed-point property. Display stays honest: this is a named decision, nothing closes, the OS contract stays [O] as the math programme. Full prose: research-contracts document, `QFT4D.OS.RECON.01` and `TFPT4D.LATTICE.ACTION.01`. Amendment (2026-08-29, review wave 4): the reviewer's precision is accepted — "a single finite box is not a world." The fundamental object is the quasilocal consistent family $\{H_\Lambda\}$ with $\tau_t = \lim e^{itH_\Lambda} A e^{-itH_\Lambda}$. Binding: continuum ($a \to 0$) optional (mathematical reinforcement); thermodynamic limit ($|\Lambda| \to \infty$) and controlled IR universality mandatory. Decision typing UNCHANGED; no [E] claim, no marker move.

Dual rest (2026-08-28; the same short form as the introduction status card). The compiler residual on this spine remains $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$. Beside it sits the strict-physical-TOE accounting of the 2026-08-27/28 contract wave

$$\begin{aligned}\text{Rest}_{\text{TOE}} = \text{SeamContinuum} &\oplus \text{4DAction} \oplus \text{ChiralMeasure} \\ &\oplus \text{MirrorGap} \oplus \text{UnitaryDynamics} \oplus \text{IR/continuum} \\ &\oplus \text{BulkSeam} \oplus \text{QuantumGravity} \\ &\oplus \text{GeneratingFunctional} \oplus \text{InitialState},\end{aligned}$$

every summand a named [O] (or standing-rule) contract — nothing claimed closed (`tfpt_research_contracts`; three new rows of wave 3: `GAUGE.DETLINE.FIXPOINT.01`, `GRAV.SPIN2.EMERGENCE.01`, `FTRANSFER.SK.RHO0.01`). The rank-8 ADE census of [`verification/v993_minimal_defect_selector.py`] lifts the architecture conditionality of v624 at lattice level; axiom typing UNCHANGED. Completeness wave (same day, no marker moves): [`verification/v994_mmst_criteria_battery.py`] executes 5/7 MMST identification criteria (character $[1, 0, 248, 0, 4124, 0, 34752]$; C6 written out in `articles/2026-08-28/psi_lambda_convergence_theorem_en.tex`); [`verification/v995_kms_subnet_rigidity.py`] kills all 33 mixing directions by finite KMS ($196608 \to 0$); [`verification/v996_gauge_grammar_kill.py`] excludes the naive α-grammar extension (64/0 hits; $U(1)$ control survives); [`verification/v997_spin2_smg_witnesses.py`] types the quadratic spin-2 witnesses and the SMG toy (gap 14 at $N = 8$). Contracts stay [O]. Review wave 4 (2026-08-29; suite 990 → 994, ledger 1166 → 1168, no marker moves): [`verification/v998_seam_modular_closure.py`] makes the C7 arithmetic shadow exact ($\Theta_{E_8} = E_4$ through q^8; net-level C7 stays cited); [`verification/v999_weak_collision_hp.py`] completes the finite dilation ladder through continuous time (open = thermodynamic field limit); [`verification/v1000_spin2_sk_witnesses.py`] executes the quadratic Hamiltonian half and the SK ρ_0 mechanism; [`verification/v1001_nu_pentagon_chain.py`] registers the pentagon-class candidate `FLAV.NUSCALE.06` [C] (no seesaw closure). The named AND of eight completeness gates is now the master contract `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}}$:= AND(T1..T8) vs $\text{TOE}_{\text{Validated}}$:= Complete AND independent holdouts) — it names the AND, it does not close it. Review wave 5 (2026-08-29; suite 994 → 996, ledger 1168 → 1171, no marker moves): [`verification/v1002_spin10_mirror_projector.py`] executes the DET16 finite-cluster hit (P_ϕ rank-one, gap 1, 45 commutators 0) and permanently excludes the number-preserving Casimir projector (census N=8:0, k=2:27, k=3:1463); `CHIRAL4D.MIRROR.DET16.01` [C], `CHIRAL4D.NOMIRROR.01` stays [O]. [`verification/v1003_nu_structural_nulls.py`] records both neutrino-texture nulls as binding constraints (`FLAV.NU.TEXTURE.MECHANISM.01` [O]: D4-odd *and* a $\sim 2{\times}10^{-4}$ even-odd cancellation; pure seam data excluded). The master scaling-limit theorem `SEAM.MMST.TYPEIII.CHARGED.01` [O] is named (one $\Psi_{\lambda,N} \to \Psi_\lambda$ theorem; KMS uniqueness given identification written out in `articles/2026-08-29/holomorphic_kms_extension_en.tex`, 13 pp). `GAUGE.DETLINE.FIXPOINT.01` has a unique fixed-volume stiffness point (contraction -0.632) but divergent thermodynamic step-scaling — the 1+1D toy has no plaquette self-interaction. Unit choice for v_{geo} is closed; dimensionless hierarchies remain predictive obligations. Review wave 6 (2026-08-30; suite 996 → 998, ledger 1171 → 1173, no marker moves): the day's structural find is the Milnor–Gray finite algebra ([`verification/v1004_milnor_frobenius_gray.py`]): classical cite (Milnor algebra of $x^2 + y^3 + z^5$ dimension 8, spectral numbers = E_8 exponents) plus new-in-corpus exact rank clock / Galois–Gray / socle-CP / integral D^4 bridge [E]-finite, with $PG(3, 2)$ interpretive [C]; the corpus identification `SEAM.MILNOR.LOCALRING.01` stays [O]. [`verification/v1005_relative_pencil_bridges.py`] keeps the Hankel and winding identities [E]-finite but flags the relative pencil as circular (carrier 5 already in the column sums of Q_2 and C; not a P2 selector; `AX.P2.01` unmoved). `GRAV.WEINBERG.WITTEN.01` [O] is the binding Weinberg–Witten firewall before any spin-2 marker move. Review wave 7 (2026-08-30; suite 998 → 1002, ledger 1173 no new rows, no marker moves): five MMST lemmas in-house ([`verification/v1006_`

mmst_lemma_battery.py]; SEAM.MMST.TYPEIII.CHARGED.01 stays [O]; residual TEL-B-EXTERNAL + ALG-EXH boxed; cited articles/2026-08-30/mmst_charged_scaling_limit_en.tex). DET16 T1/T2 proved and T3 Michalakis–Zwolak cited-verified ([verification/v1007_det16_stability_battery.py]; CHIRAL4D.MIRROR.DET16.01 stays [C]; 1/192 and BDL boxed). The 2+1D master-object scaffold is coherent ([verification/v1008_master_assembly_scaffold.py]; combinatorial $K = 0$, unique-contractive selector, QWZ wall + DET4 mirror, $\mathbb{Z}_6$ assembly; lattice-fundamental Decision and GAUGE.DETLINE.FIXPOINT.01 unmoved). ρ_0 uniqueness is finite-closed with two typed review corrections ([verification/v1009_rho0_minimizer.py]; V1000_CONSEQUENCE_MISMATCH + CENTER_FLATNESS_PREMISE_FALSE; FTRANSFER.SK.RHO0.01 stays [O]). Afternoon harvest (2026-08-30; suite 1002 → 1004, ledger 1173 no new rows, no marker moves): the simplicity campaign finds no smaller axiom core ([verification/v1010_simplicity_bridge_census.py]) — remainder minimum $S_4 = 2$, the simplicity lives in the theorems; the relative-pencil circularity flag is rehabilitated as forced provenance (ladder forces Q sums $(9, 5, 1)$; W remains conditionally unique 1078 → 1 given two directional premises); the Milnor strong all-structure bridge is finitely obstructed at $\mathbb{F}_2$ and every upstairs home (only the coarse D+pairing bridge exists; SEAM.MILNOR.LOCALRING.01 stays [O]). The 3+1D ladder is complete at scaffold level ([verification/v1011_3p1d_ladder_battery.py]): minimal object viable, fully coupled link–wall, Kronecker limitation removed; matching scale of the coupling functional must sit in the deconfined/weak regime (GAUGE.DETLINE.FIXPOINT.01 stays [O]). Evening TOE-gate wave (2026-08-30; suite 1004 → 1005, ledger 1173 no new rows, no marker moves): four executed witnesses in one battery ([verification/v1012_toe_gate_witnesses.py]) — T5 IR universality on the physical QWZ edge ($c \to 1$, clustering dichotomy, Lieb–Robinson cone, curvature → 1/6) with a typed SPLIT (gauge holonomy + seam clock are not propagating modes; common-c needs dynamical gauge excitations); T8 ρ_0 selection-principle vacuity (hard sectors response-dead, KMS-affine sets selection-vacuous; a non-entropic principle is required); T6 κ six-fold NULL completing the scaffold-level neutrino-mechanism bound (the 1.885×10^{-4} cancellation needs the true 4D functional); T7 interacting TT stress mechanism (conservation, positivity, sum rules, Z-persistence under ϕ^4; gapped, as expected for massive generic matter; missing TFPT content, $k \to 0$ volume scaling, Ward identities). TFPT.TOE.COMPLETE.01, FTRANSFER.SK.RHO0.01, FLAV.NU.TEXTURE.MECHANISM.01 and GRAV.SPIN2.EMERGENCE.01 stay [O]. Late-evening harvest (2026-08-30; suite 1005 → 1007, ledger 1173 no new rows, no marker moves): the thermodynamic-dynamics theorem closes the *mandatory* T3 dynamics leg at Hamiltonian-class level ([verification/v1013_thermodynamic_dynamics.py]) — uniform Lieb–Robinson bound $v_{\mathrm{LR}} = 32086e/225 \sim 387.64$, norm-convergent τ_t, gauge-invariant quasilocal $\mathcal{A}^G$, existence of ground/KMS states; uniqueness, the limit gap and IR universality remain open; Decision typing UNCHANGED. The finite detline restriction isomorphism is verified and W-bridge P-anch is derived ([verification/v1014_bridge_refinements.py]) — orientation fixes the conjugate branch, the residual constant phase is the A_0 reference normalization; P-dem narrows to the character-blind determinant-response map. Within the finite compiler/W-bridge lane, the remaining identification was routed through the external MMST/Quillen leg (TEL-B/ALG-EXH/P1 + that response map + Bismut–Freed/A_0). This historical reduction does not classify T3–T8. Monday-morning harvest (2026-08-31; suite 1007 → 1009, ledger 1173 no new rows, no marker moves): the axiom core is now fully premise-derived at finite level ([verification/v1015_axiom_core_closure.py]) — P-dem $r = (1, 1, 1)$ exactly, both W-bridge premises derived, remainder 0 modulo the externalized MMST identification; AX.P2.01 stays an axiom. The first viable canonical-state principle is the R_4 seam-modular cap; TEL-B narrows to the scalar dyadic increment $A_R < 43$; N-uniform CAR nuclearity is proved as compactness input; the third-route volume theorem is cited-verified for finite-group DET chains ([verification/v1016_state_gap_batteries.py]). The old "every remaining fundamental gap" wording is scope-corrected: only that finite compiler/W-bridge lane was reduced to MMST plus the 4D functional; independent T3–T8 mechanisms and a shared 3+1D parent remained open. Directed-readout and Coxeter–Euler harvest (2026-09-02; suite 1010 → 1012, ledger 1174 → 1176): [verification/v1018_e8_directed_readout.py] promotes seven exact E_8 readout cells (E8.DIRECTED.READOUT.01 [E] Identity; C7 Gauss-code transform stays open, rem:c7-audit). [verification/v1019_coxeter_euler_completion.py] is the Coxeter–Euler completion (E8.COXETER.EULER.COMPLETION.01 [E] Identity plus Numerical Euler products at $X = 10^5$; class statement: vanishing linear term generic from $\operatorname{Tr} C = -1$; Beurling-generic; fence: the trace-free completion is zero-free and pole-free in $\operatorname{Re} s > 1/2$; the splitting remains open). Round-4 TOE integration (2026-09-05; suite modules [verification/v1026_telb_round4_closure.py], [verification/v1027_signed_det_car_wall.py], [verification/v1028_gauss_d4_cap.py], [verification/v1029_local_tensor_constraints.py], [verification/v1030_alg2_sector_frame.py]): T1 retains only finite axiom-core/W-bridge support while unconditional provenance and the dimension

selector remain open; T2 closes the relaxed uniform TEL-B norm $\|R_N\|_{\mathrm{HS}} < 2.995906 < 3$ for all even $N \geq 16$, while microscopic one-edge generation, joint-adjoint tails, ALG-EXH and MMST remain open. The conditional sector/frame theorem does not undo the old Vandermonde refutation. T3 remains a Hamiltonian-class result; T4 gains exact DET-CAR modes and a signed $z = 1$ wall for fixed classical backgrounds, but no hard bare-mirror response gap, chiral quantum-link construction, measure or continuum theorem; T5 retains finite IR witnesses and the v1020 spectator obstruction but lacks prime completion/OS/confinement/scattering; T6 gains full Gauss dimensions 3 mod 5 for one site and 0 mod 5 for the open cube, but no mass texture. T7 gains a local two-mode tensor target that requires removal of its unstable global canonical zero mode, not a graviton. T8 gains a symmetric internal cap with gap one; the two-dimensional invariant space prevents a symmetry-only selector of global ρ_0. All T1–T8 markers and `TFPT.TOE.COMPLETE.01` stay [O]; no marker moves. A single shared 3+1D parent satisfying T3–T8 is still missing.

Round 7 (2026-09-05; 233/233 typed checks with the round-6 prerequisite) adds the explicit free zero-mean Weyl/Fock quantization and continuum limit ([`verification/v1031_quantum_tensor_curvature.py`]), then a positive local Lorentz-covariant curvature field carrying helicities ± 2 ([`verification/v1032_covariant_curvature.py`]). The microscopic TFPT embedding and universal nonlinear coupling are still absent. Auxiliary $\mathbb{Z}_4$ flux-register corners are constructed, but the simple quarter twist has $h = 1/4$ rather than $h_\lambda = 1$ ([`verification/v1033_charged_disorder.py`]). The source's factorized onsite mirror gap is at least $1 + \sqrt{3}/2$; the ordinary hopping bond instead violates the proposed vacuum-relative form bound, with no conclusion of interacting gap closure ([`verification/v1034_parent_mirror.py`]). The physical-Fourier linear Ward system and prescribed-source response are explicit; dynamical quantum matter and the removed homogeneous block remain binding limitations ([`verification/v1035_matter_coupling.py`]). The full proofs are included in the contracts paper. No T1–T8 or `TFPT.TOE.COMPLETE.01` marker moves.

Research consolidation (2026-09-09). The contracts paper now includes the post-Round 27 archive through the source-based charged CAR field limit. On the original width-eight QWZ strip, local smooth fields and both adjoints converge relative to the actual filled sea, with charge and finite-energy generator control. This constructs integer edge charges, not the half-charged E_8 spinor or the common TFPT parent. A separate compact-$U(1)$ test parent now has all-time quasi-local dynamics (not point-norm continuous on the full bounded algebra), neutral physical ground-state representations and nonzero Wilson-loop excitation weight. Its selection, chirality and continuum remain open. Clock/charge audits rule out a site-fixed rotor-independent internal identification, not every possible geometric one. Full statements and the next inter-sector test are in `experiments/theory-contracts/RESEARCH_2026-09-09.md`. Historical seam-net reductions below concern conditional target routes; they do not discharge the present microscopic charged-field matching. All T1–T8 and `TFPT.TOE.COMPLETE.01` remain [O].

Earlier research update through Round 27 (2026-09-07). The shipped non-RH theory contracts now extend to the original finite-regulator $g = 0$ scalar/charge model, with every integer charge and E8 phase retained. A complete charge-diagonal reference Z_D gives $e^{-48\beta NJ} Z_D \leq Z \leq Z_{\mathrm{abs}} \leq Z_D$ by all-block pinching and joint scalar/charge Hölder. The artificial extra charge heat-trace factor is removed from the sign, memory and hop error budgets, not the remaining exponential dependence on volume and hopping. At the earlier $N = 27, T = 3, \beta = 1, J = 0.1$ comparison, certified cutoffs improve from $K4096/P962$ to $K2670/P354$ for less than 1% omission error. A specified weak-hopping $L9/T33$ example retains 25 of 33 quadratic time offsets with summed memory error at most 0.2045% and combined omission error below 1%. Its full scalar determinant remains unchanged. The complete interacting partition is not evaluated; practical charge summation, interacting continuum, real-time dynamics, common-parent selection, chirality and gravity are still open. Proof, costs and reproduction are in the contracts paper and `experiments/theory-contracts/ROUND27_STATUS.md`. No status moves.

Tier B made concrete ([`verification/v277_seam_calderon_e8_match.py`]). The seam-Calderón $\to (E_8)_1$ identification is reduced to a *single* bit. [E] the full $(E_8)_1$ matching target is computed and matches: c=8 (Sugawara 248/31), $\det \mathrm{Cartan}(E_8)$=1, the holomorphic character $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$ with a single primary and exactly $248 = \dim E_8$ weight-1 currents. [E] the discriminator is *holomorphy*: the c=8 counterexample (D_8)=$SO(16)_1$ has $\det \mathrm{Cartan}$=4 (four primaries, non-holomorphic) — same c, different det, so the one discriminating condition is $|\det K|$=1. [C] the seam side matches via the flat τ=i pillowcase (v276) and the Kummer/K3 (v260). [C] the residual is exactly that single holomorphy bit — now *closed modulo cited theorems* on the MMST route `SEAM.EQUIV.MMST.01` (lattice v367/v368 + the S_3 stack v376–v379, Lean `FORM.SEAM.MMST.01`; cited continuum existence [O], v336) (`QGAMB.TIERB.01`).

The modular-data view (`[verification/v281_holomorphy_modular_data.py]`). The Tier-B bit is the anyon count: #anyons = $|\det \mathrm{Gram}|$ (E8 → 1 holomorphic, $SO(16) \to 4$), the condensation tower $D_5 \oplus A_3\,(16) \to D_8\,(4) \to E_8\,(1)$ each step a Lagrangian Z_2 boson, and Gauss–Milgram returns c=8 — so the open bit is simply "does the seam net condense to $\det K$=1" (`QGAMB.MODULAR.01`).

The live kill-test scorecard (`[verification/v283_live_killtest_scorecard.py]`). The recent round's computable predictions confronted with current data: $\sin^2\theta_{12}$ ($+0.31\sigma$), $\sin^2\theta_{13}$ ($+1.45\sigma$), J_{PMNS}=−0.0297, Σm_ν floor 0.0586 eV, proton decay τ_p=1.55×10^{35} yr — all consistent at $< 2\sigma$, with two sharp near-term kill tests (Σm_ν: DESI/CMB-S4; proton decay: Hyper-K) (`PRED.LIVE.01`). Two *named post-hoc* **[O]** candidates sit next to this board without touching the frozen record: the third-generation ladder-base pattern (the three $\sim 2\sigma$ mixing tensions $\theta_{13}/\theta_{23}/|V_{cb}|$ as one common $-\varphi_0^{\mathrm{ret}}$ suppression, `FLAV.THIRDGEN.PATTERN.01`, v467) and the splitting-ratio relation $\Delta m^2_{21}/\Delta m^2_{31} = |J_{\mathrm{PMNS}}|$ at -0.19σ (`FLAV.DM2RATIO.01`, v468) — both pre-registered with symmetric kills (JUNO / Belle II), see document 2.

Update (QBL chain, v108–v113): the boundary-net residual now carries *double* weight — the carrier choice $g = 5$ has been reduced to the *same* premise. The certified scalar kernel exists iff it pairs the two sheets, the certified channel counts the code identically (Pascal closure = channel identity), pair transport is minimally complete, and the single quasi-free kernel (rank = c: 5 carrier block, 8 seam hull) is the defining datum of the free $c = 8$ net this gate already posits. Closing the index-4 statement therefore also closes the carrier selection **[E]** (see `tfpt_1` and `tfpt_research_contracts`). `[verification/v113_quasifree_kernel.py]`

Update (free-bulk premise discharged, v160–v165): that "free c=8 net" premise is no longer free-standing. It is a fixed-point theorem (quasi-free ⇒ all higher cumulants κ_{2n}=0, v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap $(2/3)^6$ (v161), forced by μ_4-equivariance (v162); and the residual factors into the already-open A_2 net-existence (an assembly of established net constructions, **[C]**) and R_1 = the seam-realisation keystone `SEAM.EQUIV.01`, with the irreducible core $\{\pi, v_{\mathrm{geo}}\}$ a theorem (v165). So the metric-sector gate carries *zero new* independent open content. `[verification/v165_premise_a_closure.py]`

The deduction below the premise is Lean-formalised (`FORM.QGEO.02/FORM.QGEO.01`). After the residual reduces to the single seam-realisation postulate `QGEO.SYM.01` (the carrier μ_4 clock is the conformal deck of the seam; `tfpt_research_contracts`), its geometric normal form (the UNIFORM node: $z \mapsto iz$, $\sigma\rho\sigma = \rho^{-1}$, orbit→ μ_4, multiplier order-4 ⇔ $\zeta = \pm i$) and the conditional implication mark-local⇒ $\omega \circ \rho = \omega$ are machine-proved in the carrier-rigidity Lean project (AUDIT: PASS, only the three standard kernel axioms; mirrors `[verification/v177_seam_marking_kernel.py]` and `[verification/v210_mark_local_dtn.py]`). So everything *downstream* of the postulate is **[E]**; `QGEO.SYM.01` *itself* is the downstream face of the keystone `SEAM.EQUIV.01`, which is now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` (lattice + S_3 stack, Lean `FORM.SEAM.MMST.01`; only the cited continuum existence stays **[O]** — its extension leg since re-founded on peer-reviewed crossed-product theorems, the realisation input reduced to invariant level R1′, v469; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay **[O]**).

*A second, independent route (`[verification/v207_asymptotic_safety.py]`, **[O]**).* Beside the seam-net programme, the boundary data also support a **continuum** attack: an FRG truncation with an independent torsion field (the old UFE truncation) has a non-Gaussian UV fixed point — $\partial_t g = (2 + \eta_R)g$ is UV-repulsive at $g \to 0$ and damped at large g by the positive-definite torsion K^2 term, so $2 + \eta_R = 0$ has a root $g_\star > 0$ (intermediate value), with the scale-invariant axion coupling fixed and small, $y^2 = 16c_3^2 = 1/(4\pi^2) = 0.0253$. This is recorded as a *plausibility cross-check* that the QG gap is reachable from a second direction — it is **not** load-bearing and does *not* close G_{metric}; the primary, machine-checked route stays the seam-net. A third, quantitative approach to the keystone — the celestial and twistor continuum route, now named `SEAM.EQUIV.TWISTOR.01` (contract `CELEST.SEAM.01`, WP1–WP5d, both WP5d stages, plus the WP5e α, β, γ, δ_1, δ_2, ε_2 and ε_1 stages and the M1–M3 back-reaction milestones executed: the level-2 null ideal derived, its deleting operator explicit, the GNS limit state constructed, the two-interval index chain measured, split + strong additivity witnessed on the lattice (all three KLM ingredients of complete rationality; the continuum uplift is Xu's theorem, cited not claimed), the $q^{-1/3}$ prefactor + level k=1 pinned on the CFT side, the equivariant anomaly ledger exact on the twistor side (index bridge, −78 defect, rigid 32 T_3 residual, the honest a_0 refutation), the sphere-axion exchange sub-branch closed with a certificate, the collapse confirmed full-tensorially with the one cubic d-symbol door open (its $c_d = 1920 = |W(D_5)|$ fingerprint honestly typed look-elsewhere-loaded and convention-contingent, the stable core $c_d = 32 \times 60$, v513), "one level" lifted to a theorem of clock

invariance with the sector counter pinning $k = 1$ (v508/v509/v511/v513), the $O(-2)$ bulk-axion slot built with $\tilde{\lambda} = 6$ triply pinned (v514), the back-reacted Ω_N closed-form with $(2\pi i)^2$-integral lockstep periods and the lens-forced source charge $4 = |\mu_4|$ (v515), the twisted KS measure on the declared completion reading cancelling the $32\,T_3$ residual and supplying the $\psi = 64$ slice without the cubic d-channel (v516), the a_0 uplift coupled to the centre count on four scales, monodromy i (v517), the δ_1 chain decided — kill under the derived measure, the declared/derived measure tension stated honestly (v518) — the measure question since decided at probe level for the declared reading (single-valuedness derived from F-independence + the Quillen pairing under the typed premises TP-1..TP-4, v520), and the w_m normalisation since derived constructively at probe level ($1/\det_j$ = the Atiyah–Bott/zeta-determinant fixed-point factor, three independent sources, v523); WP5e proper [O], the global BCOV quantisation — narrowed to the global BCOV integral beyond the fibre zero-mode factor — the single remaining milestone) — is synthesised as a dedicated section in `tfpt_3_e8_audit_bootstrap`; it does not close `SEAM.EQUIV.01` either.

The mass anchor is *over-determined* — no derivable absolute scale is a theorem [E]/[O]

The single dimensionful input ($v_{\rm geo}$ = the reduced Planck mass $M_{\rm Pl}$) is not a hidden gap but the *unit*, and it is over-determined ([`verification/v274_scale_overdetermination.py`]). [E] every dimensionful output is a pure number $\times\, M_{\rm Pl}^d$: $\rho_\Lambda/M_{\rm Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$, $M_{\rm scal}/M_{\rm Pl} = c_3^{7/2}$, the whole spectrum — the absolute ν-scale (v272) and $v_{\rm geo}$ are the *same* anchor. [C] **two independent routes agree**: gravity gives $M_{\rm Pl} = (8\pi G)^{-1/2} = 2.4353{\times}10^{18}$ GeV, and inverting the compiler's dark-energy prediction with the measured $\rho_\Lambda^{1/4} = 2.24$ meV gives $M_{\rm Pl} = 2.438{\times}10^{18}$ GeV — the same Planck mass to 0.11% (*conditional* on the Λ-branch readout `LAMBDA.BRANCH.01`, [C]; the gravity route is unconditional — red-team [`verification/redteam/rt_F_qft4d.py`]). [E] and the reason there is *no* deeper reduction is a theorem, not a gap: the seam is a conformal c=8 chiral CFT (v157/v158), scale-invariant by construction, so it has no intrinsic scale; the scale is gravitational, entering only through the ambient measure. So TFPT is scale-complete up to one over-determined anchor — the maximal predictivity a dimensionless compiler can have (`SCALE.OVERDET.01`).

Summary

Item	Status	Honest handle / what is missing
η_B	[C]	downstream readout $= 6.1{\times}10^{-10}$ from closed $\Omega_b h^2$; leptogenesis interface, cleanest scenario $M_1{=}M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda$, $\tilde{m}_1{=}m_3/A_\Lambda$ (shared decuple, no seesaw scale)
m_p/m_e	[C]	cross-sector (QCD scale / closed m_e); typed runnable solver with ±7.5% uncertainty budget (v374), not a compiler power
Koide Q	[C]	source ratio $Q = 0.664$ near 2/3; the source→pole factor 53/54 is an *exact operator readout* of the missing $F{=}R{+}Q$ Sheet-Diamond corner [E]; the physical pole transfer ($F_{\rm pole}$) stays [C]
dark matter	[C]	determinant-line axion: $f_a{=}M_{\rm scal}/128$, $m_a \approx 23.8\,\mu$eV. *Post-inflationary* ($H_{\rm inf}/f_a{\approx}66$, isocurvature): the misalignment angle is a horizon average (*not* observable), the relic follows from f_a (in the post-inflationary window), domain-wall-safe via $N_{\rm DW}{=}|C|{=}1$ (v470/v472); the $3\pi/5$ spine value is retained as a structural rational, not a relic prediction — branch-level kill test, not a theory closure
muon a_μ	[X]	seam vertex $a_\mu^{\rm seam}{=}\delta_2/(2\pi){=}45/(524288\pi^9){=}2.879{\times}10^{-9}$; exact *value* [E], but the vertex-projection bridge is **dissolved** (dated kill fired 2026-08-04: WP25+BMW lattice-HVP consensus, experiment−SM $= 0.5\sigma$); compiler core untouched
quantum gravity	[E]/[C]	local field eq. carries no free dimensionless Newton coupling under named QFT/entanglement premises [C] ($G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$, v358/v359); $R + R^2$ heat-kernel grounded (G2); perturbative graviton unitarity established (Stelle ghost a truncation artefact, v304/v370/v380); ambient measure (G6) a [C] redundancy (v369)

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

Net (honest). Two of the five have a genuine quantitative handle that already lands on the data ($\eta_B = 6.1\times10^{-10}$ via the closed Ω_b; the axion DM candidate with a closed misalignment angle). One is a computed *near*-miss reported without tuning (Koide 0.664 vs 2/3). One is a typed runnable transfer (m_p/m_e as a cross-sector ratio, v374); quantum gravity has been sharpened — its local field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] (the full covariant $G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$ from fixed-volume entanglement equilibrium, v358/v359), its $R+R^2$ form heat-kernel grounded (G2), its physical sector gap-decoupled (G5), and *perturbative* graviton unitarity established (the Stelle ghost a truncation artefact, v304/v370/v380; the entire-form-factor graviton-exchange amplitude $e^{-p^2/M^2}/p^2$ is finite, UV-softened and tree-unitary, [`verification/v386_grav_amplitude.py`], and UV-*finite at every loop order* by power counting — the super-polynomial falloff drives the superficial degree of divergence to $-\infty$, regulated by the seam scale $M_{\rm scal}$, [`verification/v389_grav_loop_finiteness.py`], with full loop *unitarity* of the nonlocal form factor honestly the still-open point; the nonlocal action's IR projection *is* the atom-scaled $R+R^2$, $M_{\rm scal}^2/M_{\rm Pl}^2 = c_3^7$, so the local readout is its shadow, not a truncation with sawn-off legs, [`verification/v400_grav_nonlocal_action.py`]), leaving the ambient projective *measure* (G6) as a [C] redundancy (v369), not an open hole. None has been forced onto the compiler ladder — which is exactly the right call: the strong results stay strong precisely because the weak ones are not oversold.

The $F_{\rm transfer}$ functor as typed, runnable solvers. The four frontier transfers are one functor — compiler *source* → measured observable via cited standard physics — now promoted from contracts to runnable solvers with kill tests, each [C] (never a compiler power). *Koide* ([`verification/v371_ftransfer_pole.py`]): a clean *negative* — standard QED running cannot be the source→pole transfer (it pushes Q *above* 2/3; the required ϵ is negative while QED's is positive), so the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). η_B ([`verification/v372_ftransfer_boltzmann.py`]): the integrated BDP Boltzmann ODE at the frozen $M_1{=}M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda$ gives η_B within a factor ∼1.1 of 6.1×10^{-10} with no M_R dial. *Axion* ([`verification/v373_ftransfer_relic.py`]): the finite-T misalignment ODE *decides* the branch — the frozen spine angle $3\pi/5$ lands $\Omega_a h^2$ on $\Omega_{\rm DM}$ (in band) while the hilltop 170° over-produces (∼5.4×); in the natural post-inflationary scenario ($H_{\rm inf}/f_a{\approx}66$) the angle is instead a horizon average and the relic follows from f_a with $N_{\rm DW}{=}1$. m_p/m_e ([`verification/v374_ftransfer_qcd.py`]): the observed 1836.15 sits inside the propagated ±7.5% band of the two declared externals ($\alpha_s(M_Z)$, C_p), with lattice tightening as the kill. Each remains [C] — a transfer target, not a forced compiler number.

Shared source and transfer update: 3 November 2026

What this update establishes

The original structured boundary source, its common action and the $S_{\mathrm{adm}}/Z_{\mathrm{rel}}/\mathrm{OS}/\Gamma_{\mathrm{ren}}$ chain are retained. Several complete source calls, field maps and transfer kernels have now been evaluated in their stated classes. These results include conditional charge and metric selection, physical Gauss-sector sources, vacuum and ground-source reconstruction, and a complete fixed-background Clifford kernel. The primitive spatial mirror-field insertion and its coefficient $H_{\mathrm{adm}}[F,0]$ have not been evaluated by these constructions.

The reports A–L, R104–R113 and the named follow-up reports below are dated research results, not additional ledger grades. Their mathematical, numerical, software and conditional physical scopes remain separate. No claim status is promoted here. In particular, exact reproduction of the DET16 candidate is not a universal prerequisite for every physical TFPT completion.

The common source and the corrected starting point (C, F, G)

P1 is a structured reflection-positive boundary kernel, not merely the number c_3. The operational seed already supplies its local algebra, state, time, reflection and collar operator. The original source route is

$$\begin{aligned}\text{boundary/carrier/admissibility} &\longrightarrow S_{\mathrm{adm}}, Z_{\mathrm{rel}} \longrightarrow \text{full moments},\\ \text{full moments} &\longrightarrow \mathrm{OS/CAR\ representation} \longrightarrow \Gamma_{\mathrm{ren}} \longrightarrow \text{physical readouts}.\end{aligned}$$

Each arrow retains its actual hypotheses and operator domains. A particular eightfold CAR/native-clock reconstruction is an available conditional route, not a global extra axiom. Algebraic internal clocks do not require a separately postulated laboratory gate preparation.

The action types also remain distinct. The primitive relative spectral seed, the bare input S_{adm} in the functional integral, $-\log Z$, the effective action and the master functional are not interchangeable. A proper-time profile measure and a geometric field-integration measure have different roles.

A conditional CAR reader can forget an overall factor. The complete joint source contains the weights of its integrated bosonic fields, so a field-dependent factor changes their marginal and mixed responses. Its loss in a reader is not a new freely adjustable global normalization. On the same actual integration basis the existing reweighting identity recovers the relative source from its bound weight quotient. This does not license replacing that quotient by a fitted one.

The full existing chart $U = (E, NE)$ carries D and its bound D_{ref}, their actual domains, reflection, grading, antilinear charge data, Berezin measure, bosonic weights and eta factor together. Applied to a given complete source, it gives the same unnormalized source call in new coordinates. A separate Gibbs reference-state representation is optional for this direct transport. The geometric spectral-action consumer is included in this whole-operator route; a finite coefficient reader is not equated with the full Dirac operator.

For physical BRST-closed insertions the original quotient has its stated gauge-fixing independence. This does not make arbitrary off-shell functionals gauge independent or make individual ghost determinants cancel at every background.

Charge geometry and the actual clock completion (A, B)

For the four-punctured family surface,

$$H_1 = \mathbb{Z}^4/\mathbb{Z}(1,1,1,1), \qquad H^1 = \ker\left(\mathbb{Z}^4 \xrightarrow{\sum} \mathbb{Z}\right).$$

With the declared root normalization, the meridians form the A_3^* side, their differences the A_3 root side. All four meridians have the same discriminant class. A cyclic permutation of the marked

points therefore acts trivially on that discriminant; it is not translation through the four glue classes. The earlier literal fourfold delta residue pairing does not by itself define the asserted positive metric.

On the fixed additive spinor–meridian extension, the whole positive real $W(D_5) \times C_4$-invariant metric has three parameters x, a, b. Integrality of the root pairings makes them positive integers. Unimodularity gives

$$x^5 a^2 b = 1, \qquad \text{hence} \qquad x = a = b = 1.$$

Scale, orthogonality and evenness follow within this joint completion class. The fixed charge extension, symmetries and integral self-dual completion are joint premises of this theorem. The later R102 result strengthens the standard $D_{n+1} \oplus A_{n-1}$ route: a primitive integral family insertion selects its known $n = 4$ completion under the specified pairing and locality assumptions. Neither theorem selects a physical insertion by matching a dimension.

The actual current constructor also supplies the mixed consequence: for the sixteen spinor signs and twelve ordered distinct meridian pairs, all 192 simple poles land on the corresponding family root. The 64 diagonal cases give their normalized double poles, with the original cocycle and adjoint convention.

For the actual later chart M and native clock σ, the integral closure of D_8 and RD_8, $R = M^T \sigma M$, is the index-two extension $D_8 + \mathbb{Z}(q/2)$, with even integral Gram matrix and determinant one. A marked source word already produces the requisite spinor class. This is a concrete completion in the given source chart; the general clock-closure principle predates this calculation.

The marked bosonic DHR sector supplies the canonical local extension B with conditional expectation $\mathcal{E} : B \to A$ and $\omega_B = \omega_A \circ \mathcal{E}$. Neutral final words preserve their old values while charged intermediate paths are retained. The given sector covariance, square and cocycle conditions determine the time lift. A passive background transport alone does not establish common physical admission of both algebras. This special route does not create a universal new CAR, clock or endpoint-normalization requirement.

Variable fields, ordered sources and retained reference factors (D, E)

The original local source reader no longer needs constant Higgs backgrounds in its stated circle class. The smooth bounded variable one-form has the required Hilbert–Schmidt control, entire ordered $\det_2$ source, trace correction and all local response orders. The actual Galerkin calls converge to that same source operator. This is an analytic application to the existing coefficient family; it is not a probability law for the fluctuating Higgs field.

The actual Dyson source has separately been constructed. It is not replaced by the exponential of a smeared linear reader. The coherent two-face graph retains the crossing blocks and produces the positive history kernel and mixed CAR words. For the stated normal-variable graph, its polar operator fixes the transported partner word; the same untransported operator on both faces need not give the same answer. The existing unpaired echo amplitude is also evaluated with its actual ground-zero excitation convention and particle/hole signs.

The complete geometric, reference, eta and zero-source factors remain attached to their original callers. In particular the actual `v259` profile call uses its heat profile, not a Fock logarithm. No automatic general cancellation of that logarithm or of all source phases is inferred.

Physical source existence, ground words and common geometry (H, I, J)

The given finite-range uniformly local Hamiltonian class has its thermodynamic real-time norm dynamics and ground/KMS states. Restrictions of one limiting state are compatible. Independently restarted small bare Gibbs weights need not be projective. Exact elimination carries the effective fiber weights, exterior determinants, higher vertices and memory, rather than imposing the original bare ansatz again at each cut.

Gauge invariance of an ambient Gibbs density is weaker than Gauss-sector support. With the

original complete interior stars preserved by the Hamiltonian, the source

$$\rho_{\Lambda,\mathrm{phys}} = \frac{P_\Lambda e^{-\beta H_\Lambda}}{\mathrm{Tr}(P_\Lambda e^{-\beta H_\Lambda})}$$

is positive and nonempty in the stated regular-link, free-boundary class. Its weak-* limit gives physical ground/KMS sources and the actual quotient representation. Broken seam or boundary stars are not added. The projected state can differ from the earlier ambient state.

The same source supplies all bounded ordered real and Euclidean time words. The thermal reflected word form is a Hilbert–Schmidt Gram form; product-norm bounds and the local entire-source majorant pass along the same limiting subnet. For the physical ground source, all bounded Euclidean words converge to the existing positive GNS semigroup words. The whole finite ground-projector trace has a diagonal Gibbs approximation without a uniform thermodynamic gap. It does not select an arbitrary vector in a degenerate ground space.

For an applicable complete vacuum OS source,

$$q[J] = \frac{Z_{\mathrm{rel}}[J]}{Z_{\mathrm{rel}}[0]} = \frac{Z_{\mathrm{adm}}[J]}{Z_{\mathrm{adm}}[0]}, \qquad \Omega = [1], \quad \|\Omega\| = 1, \quad H\Omega = 0.$$

Only the outer source-independent reference denominator cancels here; inner reference weights and absolute energy/eta factors remain. The actual full Euclidean covariance and appropriate reconstruction regularity are needed for the invariant vacuum and forward-cone conclusion. A bounded-net route retains its time-zero hypothesis. The existing neutral local $Q = 0$ vacuum/BW construction is preserved; its parabolic generator is not the unchanged gapped circle energy.

On the original geometric Lorentz branch, whole-source covariance and an invariant vacuum carry the common metric into the physical pole geometry through the Ward identity. The complete mixed kinetic tensor must be tested, including the time–space terms. If the same source supplies its stated uniform Hermitian error bounds ϵ, η,

$$\sqrt{\eta^2 + 1 - \epsilon} - \eta \;\le\; \frac{v}{c} \;\le\; \sqrt{\eta^2 + 1 + \epsilon} + \eta, \qquad 0 \le \epsilon < 1.$$

This is a whole-class bound, not a fit of ϵ, η to the native static data. The coupled native static response, separate Wilson-slab consumer and typed velocity comparison are distinct calls. A Lieb–Robinson graph bound is not a Lorentz light cone. Thermal Hamiltonian time, vacuum time and a BW modular boost retain their actual identifications.

For the registered lattice route, the thermodynamic limit and controlled IR universality remain required. An $a \to 0$ limit is an optional strengthening there, not a universal additional gate.

The complete transfer comparison and its field requirements (K, R104–R108)

The complete original word source gives its actual pairing and time kernel,

$$G(F, G) = q((\Theta F)G), \qquad K_t(F, G) = q((\Theta F)\theta_t G).$$

Its first derivative defines the generator form on the actual quotient. Equal full pairings and closed forms, with actual autonomous sewing, determine the whole transfer. A discrete source can instead be compared at its whole fixed step. In general $Pe^{-tH}P$ is not e^{-tPHP}; a nonreducing compression can lose the required sewing.

For the actual ordinary finite thermal trace, full joint-block reset insertions give

$$C_{ij}(t) = \frac{(e^{-(\beta-t)H})_{aa}(e^{-tH})_{ij}}{Z_\beta}, \qquad p_a = \omega_\beta(E_{aa}) > 0,$$

$$H_{ij} = -\frac{C'_{ij}(0)}{p_a} \quad (i \neq j), \qquad H_{ii} - H_{jj} = -\frac{C'_{ii}(0) - C'_{jj}(0)}{p_a}.$$

The actual unnormalized positive trace fixes the common scalar. Resetting only the mirror factor requires all actual rest tests to recover a rest-valued block. A removed physical sector is not inverted, and a charged inadmissible insertion is not declared observable. These are extraction formulas; they do not constitute an evaluated primitive field insertion. The off-diagonal qualification is essential.

The supplied DET16 factor is itself completely evaluable:

$$H_{\mathrm{DET}} = 2(I - P_\varphi), \quad |\varphi\rangle = \frac{|0\rangle + e^{i\varphi}|F\rangle}{\sqrt{2}}, \qquad e^{-\beta H_{\mathrm{DET}}} = aI + bP_\varphi,$$

$$a = e^{-2\beta}, \qquad b = 1 - a, \qquad Z_{\mathrm{DET}} = 65536a + b.$$

Its equal-time covariance is $\delta_{ij}/2$, but its higher density and sixteen-field responses are not quasifree for $\beta > 0$. R104 gives a matrix-free cubic renewal evaluation of every finite ordered CAR word. The conditional separated-source reduction retains the actual gauge weights and rest source; it does not solve the entire rest lattice.

R105 isolates the exact charge-sixteen generator block $H_{F0} = -e^{i\varphi} I_R$. With mirror number bandwidth s, $\langle F|H^j|0\rangle = 0$ for $js < 16$ on the stated regulator/core. At bandwidth two the first possible order is eight, whereas the literal DET transfer begins linearly. The first possible order is not a guarantee of a nonzero coefficient. No bandwidth-two premise is asserted for the whole original S_{adm}.

The original even DET rotation carries the complete given source, gauge operators and source legs. R106 evaluates all sixteen field time laws and distinguishes the flat parent from the additive ΔN_a parent: a shared distinguished ground vector does not make their two-excitation energies equal. R107 supplies the rest-valued coefficient reader from an actual CAR field derivation, without demanding an autonomous cell or a new empty OS vacuum.

The original 96-dimensional Yukawa/Majorana field block retains all its given couplings and explicit family/antiparticle transitions. It is not a list of 96 independent couplings or a derived physical many-body mirror Hamiltonian.

K gives the exact internal $v474/v1002$ Clifford basis intertwiner, with determinant $+1$, so its exterior lift carries the top determinant without a new phase. This internal map does not supply spatial Weyl spin or a smearing/time map. The original vector source uses sixteen real Majoranas, hence eight complex modes in one packet. Native DET16 uses sixteen complex spinor annihilators. R108's specified double packet is tested by its actual central and physical-color actions, not by equal Fock dimension.

The precise common point-local Weyl assignment with scalar rest determinant coefficient is excluded: its fermionic internal epsilon contraction transforms as $\mathrm{Sym}^{16}\mathbb{C}^2$ (spin eight), not a rotation scalar. The state witness additionally requires the stated invariant fixed background and common s-wave smear. Orbital or rest spin, other field maps, larger charged sectors and effective IR descriptions are not excluded by this local theorem. The original open QWZ wall smearing remains a valid coupled field in its four-flavor scope.

The complete given Clifford field and kernel (L)

Retain the normal-independent product collar and its actual smooth classical odd coefficient:

$$B = Z \otimes A - S \otimes \Phi, \quad [A, \gamma] = 0, \quad \{\gamma, \Phi\} = 0, \qquad h = A\gamma - i\gamma\Phi.$$

For $Ev = (P_+ v, iP_- v)$, where $P_\pm = (1 \pm \gamma)/2$, $BE = Eh$ on the transported H^1 domain and $h^2 = A^2 + \Phi^2 - i[A, \Phi]$ on H^2. No commutation of A and Φ is imposed.

The oriented two-face covariance is

$$C = (1 + e^{2\ell h})^{-1}, \quad X = (2\cosh \ell h)^{-1}, \qquad P_h = \begin{pmatrix} 1 - C & X \\ X & C \end{pmatrix}.$$

The given Clifford quantization uses this full graph as its quantum covariance. This is a named source premise, not a derivation of every physical S_{adm} from a classical Cauchy projection.

On its existing thermal Hilbert–Schmidt representation, with genuine whole-source parity Γ, put $a_R = L_a$, $a_L = -\Gamma R_a$. The complete native map is

$$\iota c(T_{\text{faces}}(v_0, v_1, \bar{w}_0, \bar{w}_1)) = a_R^*(v_0) - a_L^*(v_1) + a_R(w_0) - a_L(w_1).$$

Here a is the old signed charged field: $a(f) = b(Q_+ f) + d^*(\overline{Q_- f})$ with $Q_\pm$ the spectral signs of h, distinct from the grading $P_\pm$. Creators are linear and annihilators antilinear in their tests. The opposite real-structure sign on the second face is retained. The actual product reflection is implemented by the graded HS conjugation; the even reflected form is a positive HS Gram form, with the fixed twist for odd words.

The geometric field time and source time have the same parameter. The positive excitation generator is

$$E_{\text{exc}} = h|_{Q_+} \oplus (-\overline{h|_{Q_-}}), \qquad H = d\Gamma(E_{\text{exc}}) \geq 0.$$

The cyclic HS implementation has generator $L_H - R_H$, not H. For each finite compatible regulator the whole oriented kernel is

$$Q_{\text{raw}}(s) = \bigoplus_n \Lambda^n e^{-sh}, \qquad Q_{\text{heat}}(s) = e^{sE_{\text{sea}}} Q_{\text{raw}}(s) = e^{-sH}, \quad E_{\text{sea}} = \operatorname{Tr}(\mathbf{1}_{h<0} h),$$

$$(\eta | Q_{\text{heat}}(s) | \xi) = e^{sE_{\text{sea}}} \exp(\bar{\eta} e^{-sh} \xi).$$

All entries of the raw exterior lift are the corresponding minors. In infinite dimension one uses e^{-sH} intrinsically; no separate infinite sea scalar is defined. The original scalar, reference and eta factors travel with the complete source. A general interacting caller is not declared to differ from this Gaussian fiber by a source-independent scalar.

There is also an exact obstruction to an arbitrary unitary whole-kernel identification with a separated flat DET16 factor. In the full unprojected source, heat summability gives, for $\Re z > 0$,

$$Z_{\text{free}}(z) = 2^m \prod_{\epsilon_j > 0} (1 + e^{-z\epsilon_j}) \neq 0.$$

The logarithmic product converges normally. A literal rank-one ground factor of dimension $d > 2$ and gap $\Delta > 0$ would imply

$$Z_{\text{free}}(z) = Z_R(z)\,[1 + (d-1)e^{-\Delta z}],$$

with holomorphic finite Z_R on that half-plane. The bracket vanishes at $z_* = (\log(d-1) + i\pi)/\Delta$, a contradiction. An exact positive injective step suffices by its unique selfadjoint logarithm, up to a real energy offset for normalized steps.

The comparison concerns the same oriented positive physical heat operator and sewing. It does not concern arbitrary HS face refactorizations, relative quotient traces or merely the presence of a parent term. Actual Gauss/parity projection can break zero-freeness: three equal free modes restricted to even parity have $Z = 1 + 3e^{-2\epsilon z}$. Dynamical boson integration and other interacting sources are outside the free premise. The case $d = 2$ is the ordinary one-mode exception.

Larger charged sources and the remaining actual insertion (R109, R110)

The larger integral odd spinor vertices have their explicit original spatial vectors, cocycle and all sixteen V_0 time weights. Their separately supplied collective V_c is not substituted for V_0. A concrete original vertex-mode anticommutator is nonzero between two distinct components, so those components are not directly independent elementary CAR. This does not exclude other derived smearings, nonlocal maps or effective CAR realizations.

The separate declared 64-CAR/60-CCR interaction bank has

$$H_{\text{nat}} = \Delta N_b + \sum_A (g b_A^* P_A + \bar{g} P_A^* b_A).$$

Its original tensor gives $H_{\mathrm{nat}}[F,0]=0$ as a rest operator, but its sixteenth power has a nonzero complete environment block. The actual normalized witness has coefficient $16!\,8!\,\bar{g}^{16}$, and its leading Euclidean Dyson contribution is

$$8!\,\bar{g}^{16}\left(\frac{1-e^{-\Delta\tau}}{\Delta}\right)^{16}.$$

This is the first coupling coefficient with exact Δ dependence, not the complete nonperturbative propagator. The other 48 fermions and 60 bosons are retained. The compensated open transition and a vanishing single charged expectation under an invariant full weight are compatible. Neither is silently relabeled as a primitive scalar DET term.

Dated continuation: the actual residual time form (R111)

The literal original positive-parent clause leaves its residual $R_0 = H_{\mathrm{tr,pos}}$ unspecified. Once the parent and the actual reducing physical/source map have been supplied, the original annihilating admissibility penalties vanish on their admitted range. The remaining binding equation is the whole source-time form minus the represented parent form. If one realization of the printed parent clause exists, its stated positive, common-vacuum and locality conditions also admit

$$R_s = R_0 + sH_{\mathrm{par}}, \qquad s \geq 0.$$

This comparison changes excited gaps and a fixed zero-time word's time derivative while retaining the same vacuum. It is an identifiability result about those parent hypotheses. It is not a family of different generators for one fully given source: that source's full time jet fixes the actual residual.

R111 also evaluates a marked response of the original normalized C_6 Green operator. For its declared $a > 2b > 0$, the three positive excitation rates are $-\log r_1, -\log r_2, -\log r_3$, where

$$r_1 = \frac{a-2b}{a-b}, \qquad r_2 = \frac{a-2b}{a+b}, \qquad r_3 = \frac{a-2b}{a+2b}.$$

The actual diagonal label probe has response $(2r_1^k + 2r_2^k + r_3^k)/36$ and a strictly positive two-step Hankel defect, so it is not a flat single-rate penalty. This uses the given internal transfer time; the probe is not identified as a physical field insertion or a four-dimensional mass/momentum measurement. The primitive joint spectrum remains uncomputed.

Current physical status

The common original source architecture and the calculated conditional field maps are available. The actual spatial mirror-field insertion, admissibility action and full fermion/Berezin call with dynamical bosonic weights must be identified on that same source/time/domain map. Its primitive coefficient $H_{\mathrm{adm}}[F,0]$ remains `not_computed`; a recorded JSON `null` is not the number zero.
The direct zero block of the given H_{nat}, its nonzero sixteenth-order environment transition and this uncomputed primitive quantity are three different statements. Exact DET16 is a candidate comparison, not a universal physical axiom. A different source-derived viable mechanism must be assessed on its own controlled IR route. The complete physical TFPT and chiral-mirror claims are not closed by this update.

The four-mark geometry, E_8/clock structure and existing Yukawa/flavor/alpha readouts retain their original statuses and assumptions. Shared input data give correlated consequences, not additional independent empirical confirmations. R110 consolidates the source and provenance record; a technical inventory/hash audit is not a new semantic proof of every finding, a fresh full Python/Lean/Wolfram run or physical validation.

Covariant action and its variational scope (R112)

On the existing conditional PS branch, the actual two-sided unitary orbit $U = uJuJ^{-1}$ forces the quadratic fluctuation with coefficient one in the known CCvS construction:

$$UDU^* = D + A_1 + \widetilde{A}_1 + A_2, \qquad A_2 = \widehat{u}[A_1, \widehat{u}^*], \quad \widehat{u} = JuJ^{-1}.$$

For the original right-quaternion subgroup its only nonzero correction is the Majorana block $(ww^T) \otimes M_R$ and its adjoint, where $w = q_R e_\nu - e_\nu$. The full original 96-dimensional operator, including all 72 real Yukawa and 12 real symmetric Majorana directions, is retained. The corrected operator is on the same unitary orbit; its full relative spectral action is invariant when reference, selectors, domains and eta data travel under the same admissible U. The SM first-order control has $A_2 = 0$. This fixes the existing orbit correction, not all invariant additions away from that orbit or a new physical PS gauge theory.

On the separate existing Clifford field chart, the original Green pairing gives the skew kernel M_b. If its *full off-shell* Euler operator is $M_b\Psi$, inverse variation fixes

$$S_F = \tfrac{1}{2}\Psi^T M_b \Psi + S_{\text{boundary}}, \qquad \delta_b S_F = \tfrac{1}{2}\Psi^T(\delta_b M_b)\Psi + \delta_b S_{\text{boundary}}.$$

The given open polarization recovers the whole original boundary transfer. The unintegrated zero-fermion source fixes the remaining bosonic functional if it is identified on that same chart. A stronger linear Berezin source identity likewise forces this conditional quadratic action; it is not supplied by bosonic spectral stationarity. Integrating the actual dynamical bosons retains their full cumulants and induced higher fermion vertices. Neither inverse variation nor the PS orbit identifies the entire primitive physical S_{adm}.

The actual product pairing and the Craig comparison (CRAIG20261103B)

For the existing ordinary four-dimensional product branch with the whole original internal matrix Φ, the total spin/finite real structure gives an antisymmetric Grassmann kernel on each total chiral fiber. Its mass block satisfies

$$M_{\text{mass},\epsilon}^T = -M_{\text{mass},\epsilon}, \qquad \|M_{\text{mass},\epsilon}\|_F^2 = 2\,\mathrm{Tr}_{96}\,\Phi^2, \quad \epsilon = \pm 1.$$

It is nonzero for nonzero Hermitian Φ. The isolated finite KO6 pairing cannot be substituted for this full product pairing. At the specified free massless Weyl momentum $p = (\pi/2, 0, 0, 0)$ and lattice spacing one, however, the product kernel has singular values $\pi/2$, where Craig's allowed Weyl kernel has $\sqrt{2}$. Their squared defect $\pi^2/4 - 2$ excludes the direct fixed identification and any unitary basis repair of it. A derived nonunitary blocking, another regulator or a controlled continuum comparison remains outside this exclusion. The allowed Cayley fiber map also has its nontrivial Berezin Jacobian; all source legs, higher vertices, boson weights and reference phases must travel with a genuine source map. The actual Hessian of the full original S_{adm} has not been evaluated by this product calculation.

A joint Einstein and vacuum test (CONST20261103)

Take the explicitly conditional ordinary four-dimensional Dirac product with constant selfadjoint internal mass M, an equal-rank reference on the same geometry, and the single heat profile $F(u) = m_0 e^{-u}$, $m_0 > 0$. Assume its classical relative bulk trace alone supplies the vacuum and Einstein coefficients. With

$$T = \mathrm{Tr}_N\, e^{-M^2/\chi^2} - \mathrm{Tr}_N\, e^{-M_{\text{ref}}^2/\chi^2}, \qquad -N \le T \le N,$$

the complete constant-mass dependence gives

$$V = \frac{m_0 \chi^4 T}{4\pi^2}, \qquad \overline{M}^2 = \frac{m_0 \chi^2 T}{24\pi^2},$$

$$\frac{V}{\overline{M}^4} = \frac{144\pi^2}{m_0 T} \ge \frac{144\pi^2}{m_0 N} \quad (T > 0).$$

For $N = 96$, $m_0 = 1$, the bound is approximately 14.8044, incompatible with the existing dimensionless vacuum target of order 10^{-120}. A vanishing T also removes the Einstein term. No

constant Yukawa, Majorana or scale choice repairs this class. Other profiles can separate the two relative responses; dynamical bosons, quantum contributions and boundary terms are not excluded. The actual common source must produce its volume, curvature and gauge responses together with the same bound reference. The existing alpha/vacuum relation and its horizon rewrite remain conditional correlated readouts.

Internal relaxation and physical cyclic time (GENESIS20261103)

The original family transfer has eigenvalues $1, 2/3, 1/3$. It relaxes in its two nonstationary directions and has no positive integer power equal to the identity. Under the additional identification $H_M = -\log M/\tau$, $\tau > 0$, the two positive energies are $\log(3/2)/\tau$ and $\log 3/\tau$. Their ratio is irrational by unique prime factorization, so this conditional Hamiltonian has no common exact real-time period. This is not the evaluated physical TFPT Hamiltonian. The external cyclic-universe proposal assumes commensurate energy differences and a low-entropy state; it supplies no original TFPT source or geometric collapse law. Physical recurrence requires the actual occupied energy differences to satisfy $(E_j - E_k)T \in 2\pi\mathbb{Z}$, together with geometry from that same source. Exact recurrence of the whole state also repeats every internal record. No general exclusion of cyclic TFPT cosmology follows.

Whole color projection (CRAIG20261103C). On the compatible existing CAR representation, average the three whole exterior lifts of the original color-center action. Its neutral sector has rank 21856 on the full sixteen-mode exterior space; the exterior lift of the neutral one-particle projector has rank only 16. The whole coherent transfer kernel is

$$K_{c,b}(\bar{v}, u) = \frac{1}{3}\sum_{k=0}^{2} \exp(\bar{v} z^k R_b u).$$

It retains neutral composite words with charged intermediate modes. Its higher boundary coefficients are selector effects, not newly derived primitive bulk vertices. The actual Craig sector sources must retain their respective operators, Weyl spaces and measures: averaging their operators or projectors fails the original GW and projector identities. This color-center projection is not the full SU(3) singlet projection and does not identify the other original admissibility selectors or the whole primitive quantum source.

Joint marked source and selector contacts (R113). For the same conditional whole Clifford caller, assume its actual semigroup time, justified short-time product rule, compatible finite CAR factor and Hermitian boson/rest coefficient operators. The resulting joint generator is

$$H = H_B \otimes I + \sum_{j,i} V_{ji} \otimes c_j^\dagger c_i.$$

Here each V_{ji} is the original zero-time marked operator $Q_0[h_{ji}]$, not a normalized mean. The original Fock-vacuum matrix leg fixes H_B; a traced partition function does not replace that leg. The whole selector remains outside the boson integral, with its exact commutator contacts and allowed charged intermediate words. A nonreducing selector retains complementary-sector return memory; its compressed transfer cannot be replaced by evolution under PHP. Unbounded extensions need the actual common domains and strong reduction. Regular gauge/BRST directions preserve the fermion-coordinate ideal and therefore do not supply full Grassmann translations. These conditional consequences neither identify the primitive physical quantum source nor evaluate its joint spectrum.

Source and reproducibility route

The dated report labels resolve through `docs/research/fundamental-plan/RESULTS.md` and the synthesis `docs/research/fundamental-plan/CONSOLIDATION-R110-20261103.md`. The main proof owners are:

- A and B: `experiments/theory-contracts/puncture-metric-joint-completion-20261102/PROOF.txt` and `experiments/theory-contracts/joint-source-clock-field-closure-20261102/PROOF.txt`.
- C: `experiments/theory-contracts/original-paper-source-reconciliation-20261102/AUDIT.txt`.
- D–L and R104–R111: `experiments/theory-contracts/original-local-source-variable-background-20261102/`. The exact theorem and scope owners include `VARIABLE-BACKGROUND-PROOF.txt`, `PAIRED-DYSON-PROOF.txt`, `FULL-SOURCE-CORRECTION.txt`, `REFERENCE-SOURCE-COMPLETION.txt`, `PHYSICAL-THERMAL-SOURCE.txt`, `SHARED-METRIC-SOURCE.txt`, `GROUND-SOURCE.txt`, `TRANSFER-EXTRACTION.txt`, `SEED-FIELD.txt`, `ZERO-FREE-JOINT.txt`, `ZERO-FREE-ORIGIN.txt`, `LARGER-CHARGED-SOURCE-EVALUATION.txt` and `SOURCE-GENERATOR-RESIDUAL-20261103.txt`.
- R112: `experiments/theory-contracts/original-local-source-variable-background-20261102/ADMISSIBLE-ACTION-DERIVATION-20261103.txt`.
- CRAIG20261103B: `experiments/theory-contracts/craig-original-source-bridge-20261103/ACTUAL-PRODUCT-PAIRING.txt`.
- CONST20261103: `experiments/theory-contracts/constant-triangulation-20261103/CONSTANT-TRIANGULATION.txt`.
- CRAIG20261103C: `experiments/theory-contracts/craig-original-source-bridge-20261103/COLOR-PROJECTED-KERNEL.txt`.
- R113: `experiments/theory-contracts/original-local-source-variable-background-20261102/PROJECTED-JOINT-SOURCE-GENERATOR-20261103.txt`.
- GENESIS20261103: `experiments/theory-contracts/cyclic-genesis-literature-20261103/GENESIS-CYCLIC.txt`.

The original registered constructors include:

- [`verification/v252_full_finite_triple.py`]
- [`verification/v1002_spin10_mirror_projector.py`]
- [`verification/v1027_signed_det_car_wall.py`]

Their existing scope is preserved. The source reports distinguish executed native controls from analytic arguments, given inputs and uncomputed primitive insertions. No new verification-module or ledger promotion is implied by citing them in this shared update.

Whole source evaluation and field integration: 5 November 2026

Calculated source families and the remaining primitive calculation

The complete concatenation of the same supplied source already retains its fields, selectors, reference, domains, absolute factors and time. The original C6 word weights and family-history weights have executable evaluations. The conditional R104 mirror source now has whole ordered $\Gamma(r)$-inserted word coefficients and an entire nonlinear time-dependent total-number source. These are calculations in their specified original classes. The complete primitive mixed S_{adm} field integral and its interacting physical spectrum have not been evaluated by these results. No physical ledger grade changes.

Existing producers and their actual composition

The original C6 whole-word expansion already has exact convolution over all cut positions. Its Green and inverse-square-root amplitudes retain their absolute scale; the calculation at the three original cusps has exact rational remainder bounds below 10^{-35}. The existing three-state family transfer likewise gives its complete spectral, stationary and history weights. These are different typed producers. Their existence does not identify their field or time arguments.

For the same complete source, the original whole chart and ordered reference reweighting transport every allowed mixed insertion. All intermediate selector blocks remain in a word product. The general transport, reference and time maps are therefore available results, not additional free choices required after that source is given. Identification of two different historical callers remains a separate equation.

An actually evaluated whole mirror source

Use exactly the original R104 sixteen-mode CAR factor,

$$H_m = 2(I - P_\phi), \qquad |\phi\rangle = \frac{|0\rangle + e^{i\vartheta}|F\rangle}{\sqrt{2}}, \qquad \vartheta = \pi/7.$$

Thus $E_t = e^{-2t}I + (1 - e^{-2t})P_\phi$. The original cyclic rank-one expansion evaluates an ordered word by its full Fock trace and its rank-one intermediate matrix elements. For an actual canonical boundary action $\Gamma(r)$, those matrix elements use the exterior identity

$$\langle T|\Gamma(r)|S\rangle = \det r[T, S].$$

Each CAR monomial is a partial signed map on the original bit basis. Consequently the boundary blocks are finite sums of these minors, and the term with no rank-one cut is the full finite bit/minor trace. No determinant or character is inverted. The construction therefore also retains nonzero marked coefficients when the null-source character vanishes.

These unnormalized whole-word factors can be inserted into the existing R104 Gauss sum when its actual product representation is supplied. The rest trace, original reference denominator and allowed intermediate charged words remain. This does not prove an otherwise unspecified mirror gauge embedding or a primitive P1 source identification.

Let $N = \sum_{i=0}^{15} c_i^\dagger c_i$. It commutes with every canonical $\Gamma(r)$. The span of $|0\rangle, |F\rangle$ and its complement reduce the actual operators. On that span put

$$H_2 = \begin{pmatrix} 1 & -e^{-i\vartheta} \\ -e^{i\vartheta} & 1 \end{pmatrix}, \qquad N_2 = \mathrm{diag}(0, 16), \qquad d = \det r.$$

For a piecewise constant external source h_j, interval lengths Δ_j, $\beta = \sum_j \Delta_j$ and $\chi = \sum_j h_j \Delta_j$, the entire nonlinear source is exactly

$$Z_r[h] = e^{-2\beta}\left[\det(I + e^{\chi} r) - 1 - e^{16\chi} d\right] + \mathrm{Tr}_2\left[\mathrm{diag}(1, d) \prod_j e^{-\Delta_j (H_2 - h_j N_2)}\right].$$

The displayed list is the operator product from left to right; chronological input is reversed for the usual convention of later times on the left. The two subtractions remove exactly the exterior degrees replaced by the interacting two-dimensional block. This proves the formula without a source Taylor truncation. General bounded time profiles use the same finite-dimensional ordered propagator.

For the actual center filter $P_4 = \frac{1}{4}\sum_{q^4=1} q^N$,

$$Z_4[h] = e^{-2\beta}\left(1820e^{4\chi} + 12870e^{8\chi} + 1820e^{12\chi}\right) + \mathrm{Tr}_2 \prod_j e^{-\Delta_j(H_2 - h_j N_2)}.$$

For constant h, the last term is $2e^{-\beta(1-8h)}\cosh(\beta\sqrt{1+64h^2})$. It is entire in complex h, including $h = \pm i/8$. At zero source $Z_4[0] = 1 + 16511e^{-2\beta}$. This filter is not identified with the full physical admissibility projector. The total-number family is phase blind and does not replace the independently marked charged CAR source family.

The executed controls comprise 138 central native word-trace comparisons and 105 additional controls, 20 noncentral boundary controls, and 42 whole nonlinear source comparisons with 69 controls. The source comparisons use the actual 65,536-dimensional native matrix. Their largest scaled profile discrepancy was 7.21×10^{-13}; the largest absolute discrepancy including the large number-word checks was 7.83×10^{-9}. Finite numerical checks supplement the displayed invariant-space and exterior-algebra proofs.

What the whole Higgs integration still has to evaluate

The original common affine Higgs/Yukawa sewing already gives $Z_H[J - j \mid A]$, retaining any further original Higgs-dependent vertices. Its mixed matter coefficient includes $-G_{H_2\bar{H}_2} Y_\nu \overline{Y_u}$. Here

$$G_{H_2\bar{H}_2} = \partial_{J_{H_2}} \partial_{J_{\bar{H}_2}} \log Z_H[J \mid A]\big|_{J=0}$$

is a derivative with the Higgs field integrated. A derivative of a conditional fermion determinant with respect to a prescribed background Higgs value is a different operation. The later complete radial inverse uses actual integrated source values as inputs.

One may integrate the Higgs fiber first with the complete source shift, or the fermions first with the full sourced Pfaffian/minor polynomial and then perform the original Higgs integral. Already integrated fermionic weights are not inserted as a bare Higgs fiber followed by integrating the same Yukawa fermions again. At zero Pfaffian, the unnormalized complementary coefficients remain meaningful.

The original homogeneous geometric residue constructor already contains Higgs kinetic, quadratic and quartic terms and can carry damping. It is an existing component to retain, not an independently chosen replacement for the entire original weight. The coherent Gaussian resolution of identity and the volume of a fixed spacetime metric are likewise not interchangeable with the whole Higgs configuration law.

Whole configuration classes and the exact limit of the heat factor

The earlier fixed-rank bounded-profile test excludes absolute normalization over the explicitly tested flat noncompact $\mathbb{C}^2$ Higgs volume with any nonzero finite polynomial Pfaffian coefficient. The original constant-circle positive-Laplace-profile result makes its bound uniform in the mode cutoff. These scoped results are reused. They are not exclusions of every original measure or fluctuating field.

The whole already evaluated sea-corrected coordinate Fock heat factor can now be bounded in the same constant-circle class. With the actual $\alpha \in \{1/4, 3/4\}$, $t > 0$, and singular values $\sigma_a(H)$ of the complete coupled block, including Majorana,

$$F_{\alpha,t}(H) = \prod_{n\in\mathbb{Z}} \prod_{a=1}^{48} \left(1 + e^{-t\sqrt{(n+\alpha)^2 + \sigma_a(H)^2}}\right)^2.$$

All Higgs directions and all original Yukawa and Majorana ranks obey

$$1 \le F_{\alpha,t}(H) \le C_\alpha(t), \qquad C_\alpha(t) = \prod_{n\in\mathbb{Z}} (1 + e^{-t|n+\alpha|})^{96} < \infty.$$

Indeed $\log(1+x) \le x$ gives a convergent geometric majorant. For $-M \le n \le M$, the positive logarithmic remainder is uniformly at most

$$96\,\frac{e^{-t(M+1+\alpha)} + e^{-t(M+1-\alpha)}}{1 - e^{-t}}.$$

On the original S reference branch the heat quotient is in $[C_\alpha^{-1}, 1]$; on T it equals one. At the original elliptic $t = 4\pi$, both quarters give $C_\alpha \simeq 58.5078959761$. This approximation uses high-precision floating arithmetic; the displayed tail bound is analytic. All 33 targeted controls passed, including 18 native full-96 comparisons; their largest logarithmic discrepancy was 3.56×10^{-15}.

Thus this infinite heat factor alone cannot normalize the already tested flat Higgs volume. The remaining actual whole source factor, physical selector, eta data, other fields and reference are retained. This is no divergence theorem for the primitive source. Nonconstant Higgs fields, actual projected kernels and other genuine renormalized weights are outside this particular exclusion. A universal scan of all imaginable theories is neither performed nor implied by these complete statements about specified mathematical classes.

Conditional operator routes for the next evaluation. The follow-up uses the original joint R113 contact branch, rather than another fixed-background evaluation. In that branch let $\Phi = D_M + \sum_a V_a X_a$, $T_a = -i\gamma V_a$, and $A = \operatorname{Tr}_F D_Y^2 > 0$, with all original Yukawa matrices, including the Dirac-neutrino term. The existing full internal Gram identity and CAR give

$$X_a = A^{-1} \sum_{ij} (T_a)_{ij} \{c_j, [H_{\text{joint}}, c_i^\dagger]\}, \qquad U = \sum_a X_a^2.$$

For the Hamiltonian insertion $Y_{\text{ham}} = \sum_a X_a\, d\Gamma(T_a)$, the whole mixed word is a specified sum of at most eight CAR letters with three original-time derivatives. The original composite, endpoint-product and domain prescriptions are retained. The Euclidean fermion insertion keeps its own pairing and any half-factor. The actual whole neutral word precedes physical compression; charged intermediate returns are retained. All 66 targeted original-constructor controls passed. Applying this identity to the already evaluated classical-background kernel gives $X_a = x_a(H)I$ and a connected UY_{ham} response equal to zero. It does not integrate the Higgs field.

On the actual compatible finite joint tensor chart, the original selector also defines $M_P(X) = d^{-1} \operatorname{Tr}_F[P(X \otimes I)P]$. Its kernel is exactly the kernel of physical body compression. Consequently its canonical pseudoinverse recovers the same compressed body from the actual joint short-time jet. The pseudoinverse is not in general completely positive and is not a new ambient Hamiltonian. Full physical evolution follows from the compressed generator when the original P reduces the actual whole generator; otherwise the original return and memory blocks remain necessary. These are concrete conditional reconstruction routes. The primitive short-time jet, mixed field value and common interacting spectrum remain unevaluated.

Durable result route. The research records `WEIGHTS20261105B`, `SEWING20261105C`, `FIELDINT20261105D`, `HYP20261105E` and `HYP20261105F` distinguish recovered original results, executed calculations, analytic class statements and the unevaluated primitive integral. The full original source remains the common calculation target. These records neither choose a new state or measure nor change the existing c3/P1/P2, compiler, flavor, clock or physical completion ledger markers.

Appendix: computational verification

These are the *frontier* items: most carry [O] (open) or [C] (conditional) and are therefore *not* part of the pass/fail suite. The one quantitative handle that is reproduced is $\Omega_b{=}(1{-}\frac{1}{4\pi})\varphi_0^{\rm ret}$ (whence the η_B order) in `verification/v7_gravity_cosmo.py`. The $R+R^2$ structure and the gap-decoupling of the physical QG sector *are* machine-checked (`v28_gravity_fR.py`, `v36_spectral_action_g2.py`); the proton/electron ratio is now a typed runnable [C] transfer (v374) and the ambient projective limit (G6) a [C] redundancy (v369) — both conditional, not bare [O]. The compiler/SM identities they build on are checked by the suite in `verification/` (run `python verification/run_all.py`; see `verification/README.md`).

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

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- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
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